



Introduction to Emerging Technologies (CoSc1012)

Compiled By Faculty of Informatics
Course Coordinators



Chapter 1: Introduction to Emerging Technologies



Chapter Outline

- **Evolution of Technologies**
 - Introduction to the Industrial Revolution (IR)
 - The Most Important Inventions of the Industrial Revolution
 - Historical Background (IR 1.0, IR 2.0, IR 3.0)
- **Role of Data for Emerging Technologies**
- **Enabling devices and network (Programmable devices)**
 - List of some Programmable devices
- **Human to Machine Interaction**
 - Disciplines Contributing to Human-Computer Interaction (HCI)
- **Future Trends in Emerging Technologies**
 - Emerging technology trends in 2019
 - Some emerging technologies that will shape the future of you and your business



Evolution of Technologies

- **Emerging technology** is a term generally used to describe a new technology, but it may also refer to the continuing development of existing technology;
- It can have slightly different meanings when used in different areas, such as media, business, science, or education.
- The term commonly refers to **technologies that are currently developing**, or that are expected to be available within the next five to ten years, and is usually reserved for technologies that are creating or are expected to create significant social or economic effects.
- **Technological evolution** is a theory of radical transformation of society through technological development.



Evolution of Technologies(cont..)

What is the root word of technology and evolution?

- **Technology**: 1610s, “discourse or *treatise* on an art or the arts," from Greek **tekhnologia**” systematic treatment of an art, craft, or technique," originally referring to grammar, from **tekhno-** (see techno-) + -logy.
- **Evolution**: evolution means the process of developing by gradual changes.



Evolution of Technologies(cont..)

- **List of some currently available emerged technologies**
 - Artificial Intelligence
 - Blockchain
 - Augmented Reality and Virtual Reality
 - Cloud Computing
 - Angular and React
 - DevOps
 - Internet of Things (IoT)
 - Intelligent Apps (I-Apps)
 - Big Data
 - Robotic Processor Automation (RPA)



Introduction to the Industrial Revolution (IR)

- The **Industrial Revolution** was a period of major industrialization and innovation that took place during the late 1700s and early 1800s.
- An Industrial Revolution at its core occurs when a society shifts from using tools to make products to use new sources of energy, such as coal, to power machines in factories.
- The revolution started in **England**, with a series of innovations to make **labor** more **efficient** and **productive**.
- The Industrial Revolution was a time when the **manufacturing of goods** moved from **small shops and homes** to large factories.
- This shift brought about changes in culture as people moved from rural areas to big cities in order to work.



Introduction to the Industrial Revolution (IR)

- The **American Industrial Revolution** commonly referred to as the **Second Industrial Revolution**, started sometime between 1820 and 1870.
- The impact of changing the way items was manufactured had a wide reach.
- Industries such as textile manufacturing, mining, glass making, and agriculture all had undergone changes.
- From the first industrial revolution (mechanization through water and steam power) to the mass production and assembly lines using electricity in the second, the fourth industrial revolution will take what was started in the third with the adoption of computers and automation and enhance it with smart and autonomous systems fueled by data and machine learning.
- Generally, the following industrial revolutions fundamentally changed and transferred the world around us into modern society.
 - The steam engine,
 - The age of science and mass production, and
 - The rise of digital technology
 - Smart and autonomous systems fueled by data and machine learning.



The Most Important Inventions of the Industrial Revolution

- **Transportation:** The Steam Engine, The Railroad, The Diesel Engine, The Airplane.
- **Communication:** The Telegraph. The Transatlantic Cable. The Phonograph. The Telephone.
- **Industry:** The Cotton Gin. The Sewing Machine. Electric Lights.



Historical Background of IR

- The industrial revolution began in Great Britain in the late 1770s before spreading to the rest of Europe.
- The first European countries to be industrialized after England were Belgium, France, and the German states.
- The final cause of the Industrial Revolution was the effects created by the Agricultural Revolution.
- As previously stated, the Industrial Revolution began in Britain in the 18th century due in part to an increase in food production, which was the key outcome of the Agricultural Revolution.
- The four types of industries are:
 - The **primary industry** involves getting raw materials e.g. mining, farming, and fishing.
 - The **secondary industry** involves manufacturing e.g. making cars and steel.
 - **Tertiary industries** provide a service e.g. teaching and nursing.
 - The **quaternary industry** involves research and development industries e.g. IT.



Industrial Revolution (IR 1.0)

- The Industrial Revolution (IR) is described as a **transition to new manufacturing processes**.
- IR was first coined in the 1760s, during the time where this revolution began.
- The transitions in the **first IR**(IR 1.0) included
 - going from hand production methods to machines,
 - the increasing use of steam power (see Figure below),
 - the development of machine tools and the rise of the factory system.





Industrial Revolution (IR 2.0)

- IR 2.0, also known as the **Technological Revolution**, began somewhere in the 1870s.
- The advancements in IR 2.0 included
 - the development of **methods for manufacturing** interchangeable parts and
 - widespread adoption of pre-existing technological systems such as **telegraph** and **railroad networks**. This adoption allowed the vast movement of people and ideas, enhancing communication.
 - Moreover, new technological systems were introduced, such as electrical power (see Figure below) and telephones.





Industrial Revolution (IR 3.0)

- IR 3.0 introduced the transition from **mechanical** and **analog electronic** technology to **digital electronics** (see Figure on the next slide) which began from the late 1950s.
- Due to the shift towards **digitalization**, IR 3.0 was given the nickname, “**Digital Revolution**”.
- The core factor of this revolution is the **mass production** and widespread use of **digital logic circuits** and its derived technologies such as the **computer**, **handphones** and the **Internet**.
- These technological innovations have arguably transformed traditional production and business techniques enabling people to communicate with another without the need of being physically present.
- Certain practices that were enabled during IR 3.0 is still being practiced until this current day, for example – the proliferation of digital computers and digital record.



Industrial Revolution (IR 3.0)





Industrial Revolution (IR 4.0)

- Now, with advancements in various technologies such as **robotics, Internet of Things, additive manufacturing and autonomous vehicles**, IR 4.0 was coined by Klaus Schwab, in the year 2016.
- The technologies mentioned above are what you call – **cyber-physical systems**.
- A **cyber-physical system** is a mechanism that is controlled or monitored by **computer-based algorithms**, tightly integrated with the Internet and its users.



Industrial Revolution (IR 4.0)

- One example that is being widely practiced in industries today is the usage of Computer Numerical Control (CNC) machines.
- These machines are operated by giving it instructions using a computer.
- Another major breakthrough that is associated with IR 4.0 is the adoption of Artificial Intelligence (AI),
- AI is also one of the main elements that give life to Autonomous Vehicles and Automated Robots.

Industrial Revolution (IR 4.0)



Anybody Connected device (ABCD)



Role of Data for Emerging Technologies

- **Data** is regarded as the **new oil** and **strategic asset** since we are living in the age of big data, and drives or even determines the future of science, technology, the economy, and possibly everything in our world today and tomorrow.
- Data have not only triggered tremendous hype and buzz but more importantly, presents enormous challenges that in turn bring incredible innovation and economic opportunities.
- This reshaping and paradigm-shifting are driven not just by data itself but all other aspects that could be created, transformed, and/or adjusted by understanding, exploring, and utilizing data.
- The preceding trend and its potential have triggered new debate about data-intensive scientific discovery as an emerging technology, the so-called “fourth industrial revolution,”



Role of Data for Emerging Technologies

- There is no doubt, nevertheless, that the potential of data science and analytics to enable data-driven theory, economy, and professional development is increasingly being recognized.
- This involves not only core disciplines such as computing, informatics, and statistics, but also the broad-based fields of business, social science, and health/medical science.



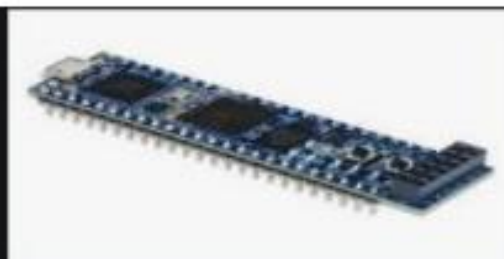
Enabling devices and network (Programmable devices)

- In the world of digital electronic systems, there are **four** basic kinds of devices: **memory**, **microprocessors**, **logic**, and **networks**.
- **Memory** devices store random information such as the contents of a spreadsheet or database.
- **Microprocessors** execute software instructions to perform a wide variety of tasks such as running a word processing program or video game.
- **Logic** devices provide specific functions, including device-to-device interfacing, data communication, signal processing, data display, timing and control operations, and almost every other function a system must perform.
- The **network** is a collection of computers, servers, mainframes, network devices, peripherals, or other devices connected to one another to allow the sharing of data.
- An excellent example of a network is the **Internet**, which
- connects millions of people all over the world

- **Programmable devices** usually refer to chips that incorporate field programmable logic devices (**FPGAs**), complex programmable logic devices (**CPLD**) and programmable logic devices (**PLD**).
- There are also devices that are the analog equivalent of these called **field-programmable analog arrays**.



Programmable Logic Device, ...
indiamart.com



Cmod A7: Breadboardable Art...
pinterest.com



Programmable Logic Device a...
dir.indiamart.com



DLP-HS-FPGA3 DLP Design [...]
mouser.com



IET & BBC Make It Digital: FR...
allaboutstem.co.uk



Styx Zynq 7020 FPGA Module...
numato.com





Enabling devices and network (Programmable devices)

- Why is a computer referred to as a programmable device?
- Because what makes a computer a computer is that **it follows a set of instructions.**
- Many electronic devices are computers that perform only one operation, but they are still following instructions that reside permanently in the unit.



List of some Programmable devices

- Achronix Speedster SPD60
- Actel's
- Altera Stratix IV GT and Arria II GX
- Atmel's AT91CAP7L
- Cypress Semiconductor's programmable system-on-chip (PSoC) family
- Lattice Semiconductor's ECP3
- Lime Microsystems' LMS6002
- Silicon Blue Technologies
- Xilinx Virtex 6 and Spartan 6
- Xmos Semiconductor L series



List of some Programmable devices

- A full range of network-related equipment referred to as **Service Enabling Devices** (SEDs), which can include:
 - Traditional channel service unit (CSU) and data service unit (DSU)
 - Modems
 - Routers
 - Switches
 - Conferencing equipment
 - Network appliances (NIDs and SIDs)
 - Hosting equipment and servers



Human to Machine Interaction

- **Human-machine interaction** (HMI) refers to the communication and interaction between a human and a machine via a user interface.
- Nowadays, natural user interfaces such as **gestures** have gained increasing attention as they allow humans to control machines through natural and intuitive behaviors
- **What is interaction in human-computer interaction?**
- HCI (human-computer interaction) is the study of how people interact with computers and to what extent computers are or are not developed for successful interaction with human beings.
- As its name implies, HCI consists of **three** parts: the **user**, the **computer** itself, and the **ways they work together**.



Human to Machine Interaction

- **How do users interact with computers?**
- The user interacts directly with hardware for the human input and output such as displays, e.g. through a graphical user interface.
- The user interacts with the computer over this software interface using the given input and output (I/O) hardware.
- **How important is human-computer interaction?**
- The goal of HCI is to improve the interaction between users and computers by making computers more user-friendly and receptive to the user's needs.
- The main advantages of HCI are simplicity, ease of deployment & operations and cost savings for smaller set-ups.
- They also reduce solution design time and integration complexity.



Disciplines Contributing to (HCI)

Cognitive psychology: Limitations, information processing, performance prediction, cooperative working, and capabilities.

Computer science: Including graphics, technology, prototyping tools, user interface management systems.

Linguistics.

Engineering and design.

Artificial intelligence.

Human factors.



Future Trends in Emerging Technologies

- **Emerging technology trends in 2019**
 - 5G Networks
 - Artificial Intelligence (AI)
 - Autonomous Devices
 - Blockchain
 - Augmented Analytics
 - Digital Twins
 - Enhanced Edge Computing and
 - Immersive Experiences in Smart Spaces

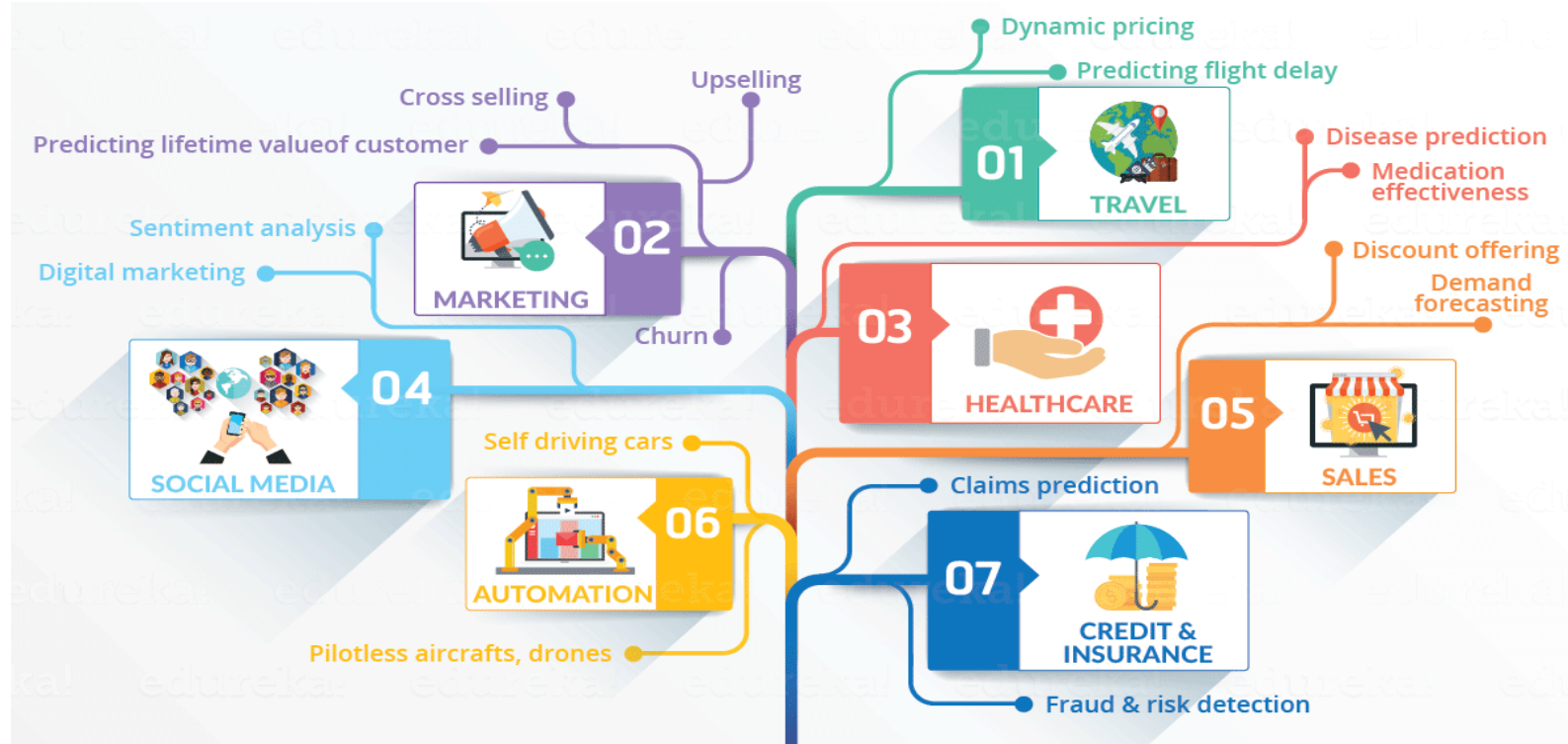


Some emerging technologies that will shape the future of you and your business

- The future is now or so they say. So-called emerging technologies are taking over our minds more and more each day.
- These are very high-level emerging technologies though.
- They sound like tools that will only affect the top tier of technology companies who employ the world's top 1% of geniuses.
- This is totally wrong. **Chatbots, virtual/augmented reality, blockchain, Ephemeral Apps and Artificial Intelligence** are already shaping your life whether you like it or not.
- At the end of the day, you can either adapt or die.



End of Chapter 1



Chapter 2: Data Science



Chapter Objective

At the end of this chapter student should be able to :

- Describe what data science is and the role of data scientists.
- Differentiate data and information.
- Describe data processing life cycle .
- Understand different data types from diverse perspectives.
- Describe data value chain in emerging era of big data.
- Understand the basics of Big Data.
- Describe the purpose of the Hadoop ecosystem components.



Chapter Outline

- Overview of data science
- Data Processing Cycle
- Data types and their representation
- Data value Chain
- Basic concepts of big data



Overview Data science

What is data science?

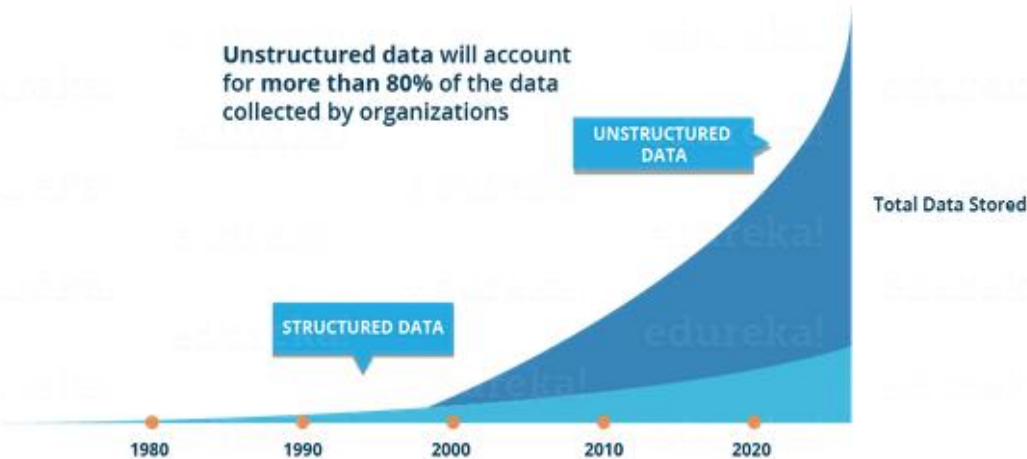
- It is a multi-disciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured, semi-structured and unstructured data.
- Data science is much more than simply analyzing data.
- It is used to create data-centric artifacts and applications that address specific **scientific, socio-political, business, or other issues**.
- It offers a range of roles and requires a range of skills.
 - Mathematics
 - Statistics
 - Data engineering
 - Visualization
 - Advanced computing





Why Data science?

Unstructured data will account for more than 80% of the data collected by organizations



- Simple tools are not capable of processing this huge volume and **variety of data**.
- The ability to take data—to be able to understand it, to process it, to extract value from it, to visualize it, to communicate it—that's going to be a hugely important skill, not only at the professional level but even at the educational level for elementary school kids, for high school kids, for college kids.

- **Data** are available in various form (structure & unstructured) and these days generated in bulk from different source , essentially free and ubiquitous data.
- **The granularity, size and accessibility** data, comprising both physical, social, commercial and political spheres has exploded in the last ' ' ,



That is Why???

Three components of Data Science:

Data science consists of three components, that is, organizing, packaging and delivering data (OPD of data).



1. Organizing the data:

Organizing is where the planning and execution of the physical storage and structure of the data takes place after applying the best practices in data handling.

2. Packaging the data:

Packaging is where the prototypes are created, the statistics is applied and the visualisation is developed. It involves logically as well as aesthetically modifying and combining the data in a presentable form.

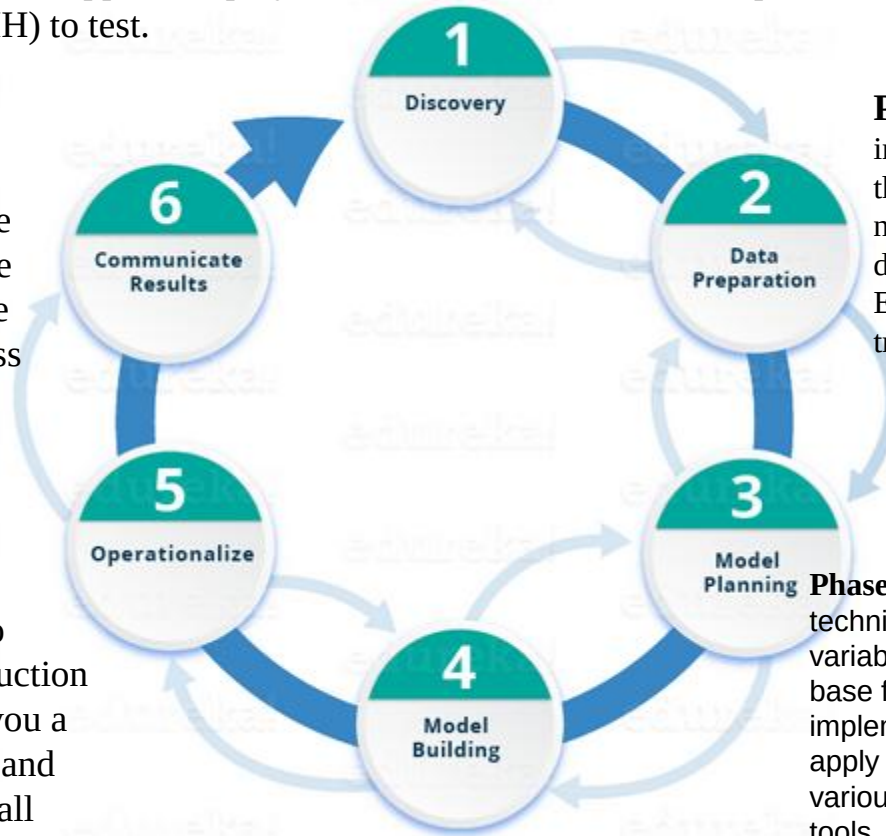
3. Delivering the data:

Delivering is where the story is narrated and the value is received. It makes sure that the final outcome has been delivered to the concerned people.



Lifecycle of Data Science

Phase 1: understand the various specifications, requirements, priorities and required budget. Ask the right questions : about the required resources present in terms of people, technology, time and data to support the project. need to frame the business problem and formulate initial hypotheses (IH) to test.



Phase 2: you require analytical sandbox in which you can perform analytics for the entire duration of the project. You need to explore, preprocess and condition data prior to modeling. You will perform ETLT (extract, transform, load and transform) to get data into the sandbox.



Phase 3 determine the methods and techniques to draw the relationships between variables. These relationships will set the base for the algorithms which you will implement in the next phase. apply Exploratory Data Analytics (EDA) using various statistical formulas and visualization tools.

Phase 4: develop datasets for training and testing purposes. Decide whether your existing tools will suffice for running the models or it will need a more robust (processing). analyze various learning techniques clustering to build the model.

Common Tools For Model Building





Data science Work flow

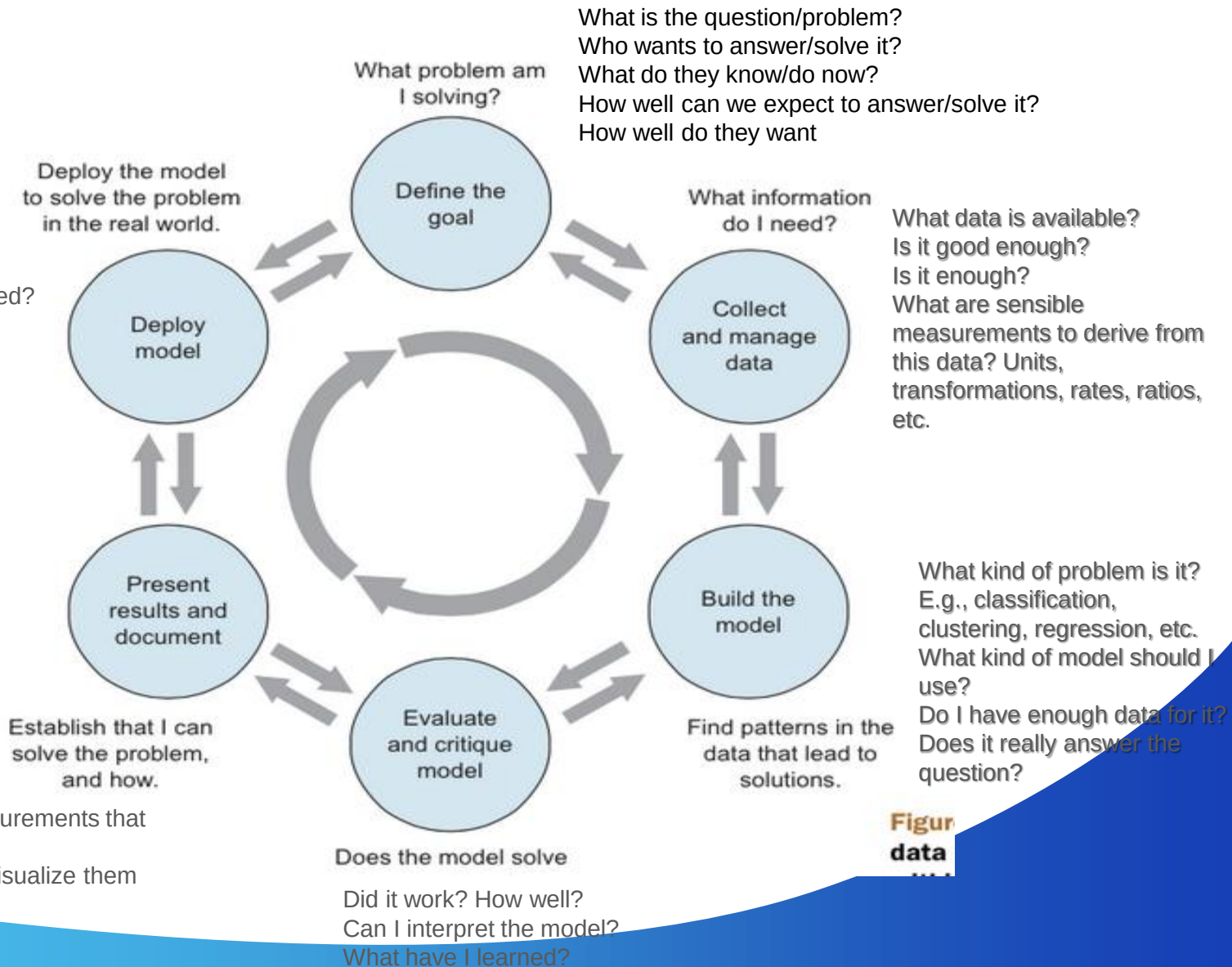
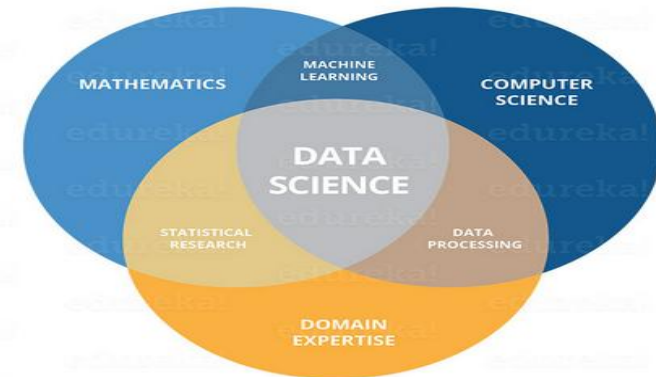


Figure
data



Who is Data Scientists ??

- A Data Scientists in simple words is one who **practices** the art of Data Science
- Data scientists are those who **crack complex data problems** with their strong expertise in **certain** scientific disciplines.
- They work with several elements related to mathematics, statistics, computer science and their **domain** (though they may not be an expert in all these fields).
- **Data Scientist should be very strong in any of these skills**





What is data?

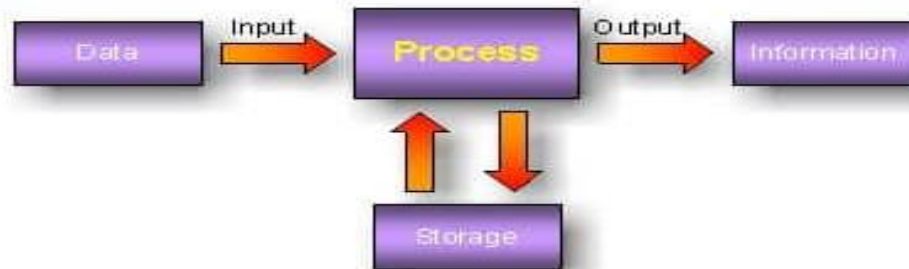
- Data can be defined as a representation of facts, concepts, or instructions in a formalized manner, which should be suitable for communication, interpretation, or processing, by human or electronic machines.
- It is **unprocessed** facts and figures represented with the help of characters such as alphabets (A-Z, a-z), digits (0-9) or special characters (+, -, /, *, , =, etc.) and picture ,sound and video.
- **Information** is the processed data on which decisions and actions are based





Data Processing Cycle

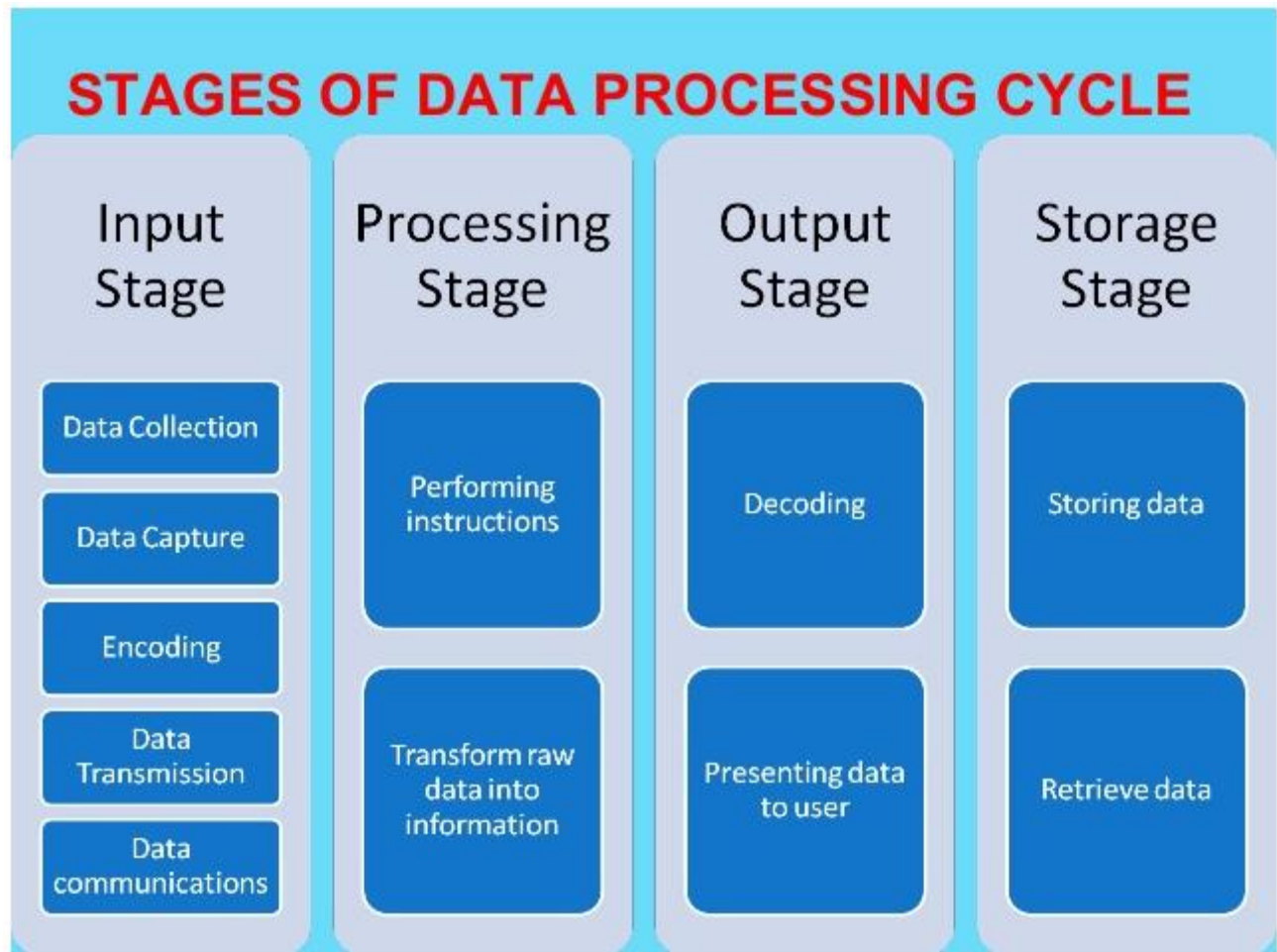
- The collection and manipulation of items of data to produce meaningful information
- Data processing is the re-structuring or re-ordering data to increase its usability
- Data processing cycle is a sequence of steps or operations for processing data to make it usable format



- Collection of raw data needs to be fed in the cycle for processing.
- The first step and called input
- Manual , electronic data processing, mechanical processing or automated means
- Outcome.
- Processed & meaning full data
- the data is useful and provides information.



Data Processing Cycle





Data types and their representation

- Data can be available in different format and can be described from different perspectives.
- **Data type** is simply an attribute of data that tells the compiler or interpreter how the programmer intends to use the data.
- **Metadata** (Data about Data) is a data that provide additional information about a specific set of data.
 - It is not a separate data structure.
- Data types can be categorized from programming perspective as follows:
 - **Integers(int)**- is used to store whole numbers, mathematically known as integers
 - **Booleans(bool)**- is used to represent restricted to one of two values: true or false
 - **Characters(char)**- is used to store a single character
 - **Floating-point numbers(float)**- is used to store real numbers
 - **Alphanumeric strings(string)**- used to store a combination of characters and numbers



Continue

There are three categories of data types from analytics perspective.

- **Structured data** – Relational data.

Name	Sex	Age	Result	Status
Abebe	M	24	90	Pass
Almaz	F	22	93	Pass

- **Semi Structured data** – XML data.

```
<student><name> Abebe</name>
<sex>Male</sex>
<age>24</age>
<Result>90</Result>
<Status>Pass</Status></student>
```

```
<student><name> Abebe</name>
<sex>Male</sex>
<age>24</age>
<Result>90</Result>
<Status>Pass</Status></student>
```

- **Unstructured data** – Word, PDF, Text, Media

- Abebe is 24 years old male student who has passed the exam by scoring 90 .
- Almas is 22 years old female student who has passed the exam by scoring 93.



Data value Chain

- Data Value Chain is a series of activities that introduce information flow within a big data system needed to generate value and useful insights from data.
- The Big Data Value Chain identifies the following key high-level activities:

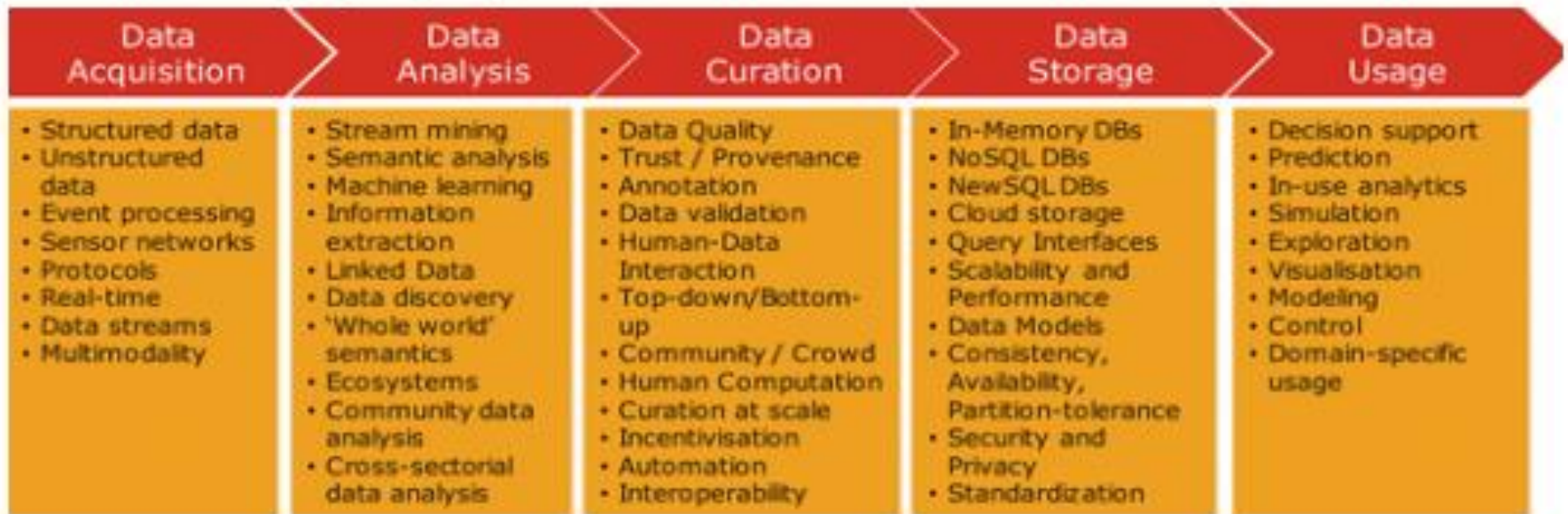


Figure 2.3 Data Value Chain

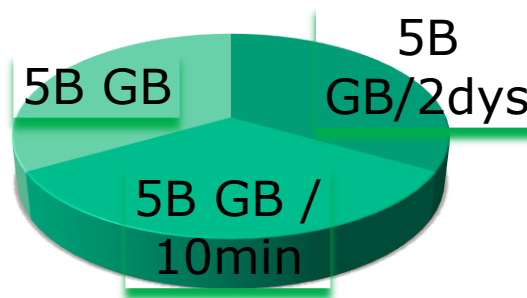


Data value Chain

- Due to the advent of new technologies, devices, and communication means like social networking sites, IoT and soon the amount of data produced by mankind is growing rapidly every year.

Data produced

■ Before 2003 ■ In 2011 ■ In 2013 ■



The amount of data produced by us

If this data is stored inside disks and pile up them, it may fill an entire football field

What Is Big Data?

- Big data is a large and complex collection of data sets that is difficult to process using on-hand database management tools or traditional data processing applications.

Sources of Big data.



Stock Exchange Data

Black Box Data

Characteristics of big data

- Very difficult to process and store it in the traditional method of data storage and processing with one computer.
- It is also characterized by 3V and more
 - Volume: large amounts of data Zeta bytes/Massive datasets
 - Velocity: Data is live streaming or in motion
 - Variety: data comes in many different forms from diverse sources
 - Veracity: can

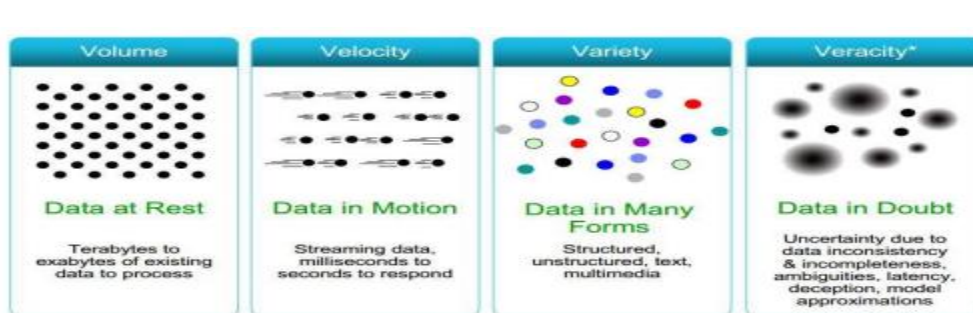
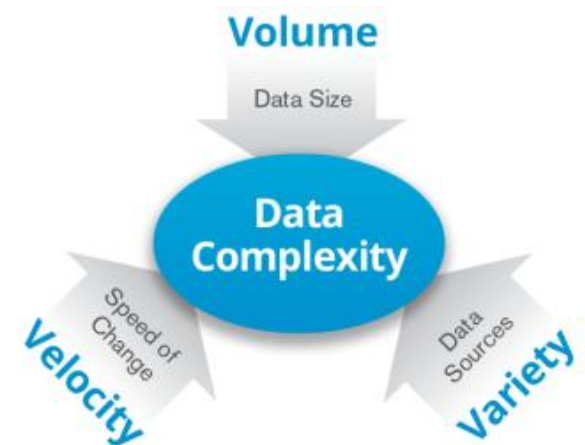


Figure 2.4 Characteristics of big data



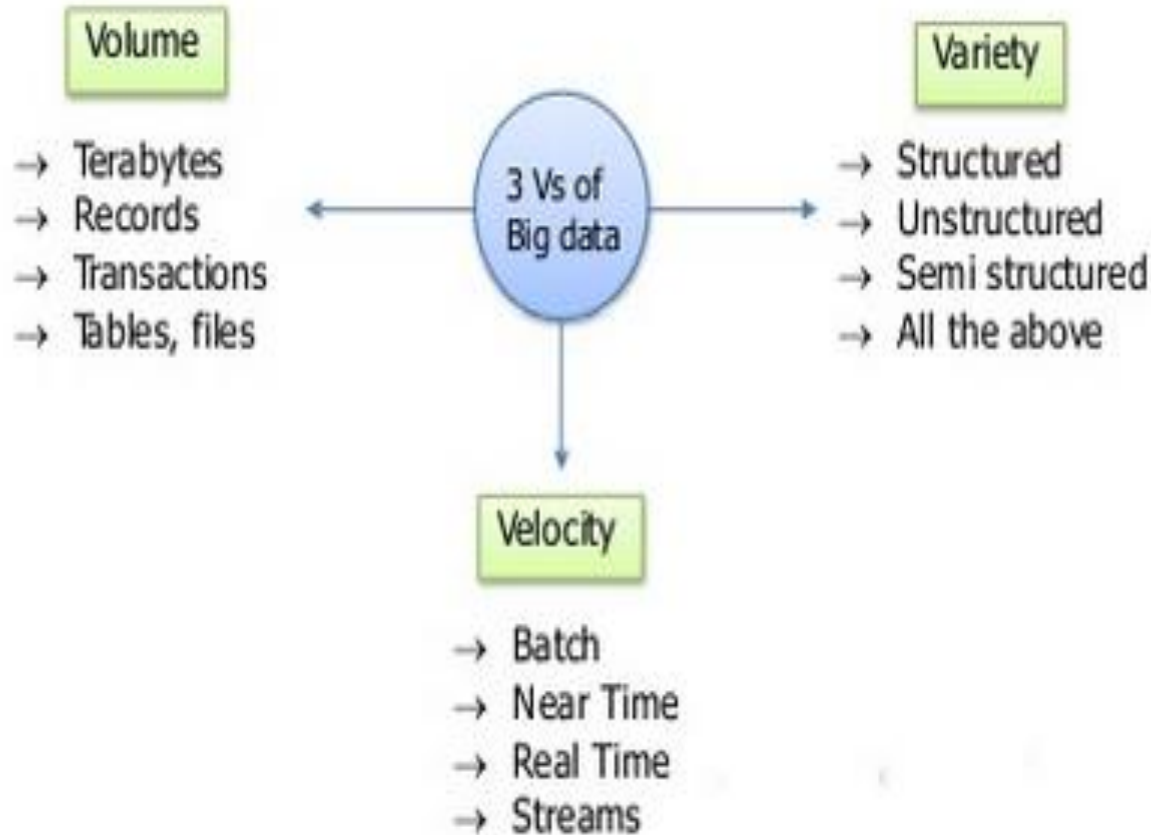
- It use cluster computing and distributed storage
 - Storage and processing is done on multiple computers



Characteristics of big data

IBM's Definition – Big Data Characteristics

<http://www-01.ibm.com/software/data/bigdata/>





Benefits of Big Data

- **Resource Pooling:** Combining the available storage space in different machine to hold data is a clear benefit, but CPU and memory pooling are also extremely important. Processing large datasets requires large amounts of all three of these resources.
- **High Availability:** Clusters can provide varying levels of fault tolerance and availability guarantees to prevent hardware or software failures from affecting access to data and processing. This becomes increasingly important as we continue to emphasize the importance of real-time analytics. 30 •
- **Easy :Scalability** Clusters make it easy to scale horizontally by adding additional machines to the group. This means the system can react to changes in resource requirements without expanding the physical resources on a machine.



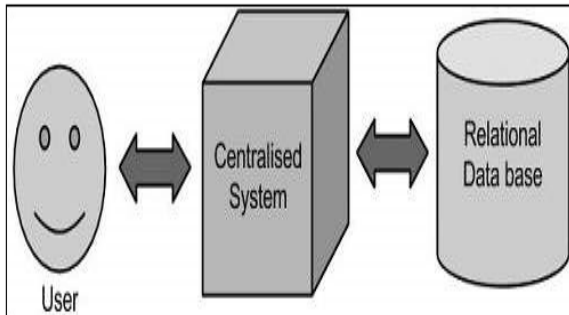
Big Data Technologies

Big data technologies are important in providing

- Helps to analyze huge data which are available in various form and stored in distributed machines.
- It gives more accurate analysis that will lead to more concrete decision-making resulting.
- It has greater operational efficiencies,
- It reduced storage and processing cost
- It reduced risks for the business by using distributed computing and storage.
- Doest require powerful and special machine for data processing and storage
- It only require an infrastructure that can manage and process huge volumes different data type
 - structured and
 - unstructured data in real time and
 - can protect data privacy and security.
- There are various technologies in the market from different vendors including Amazon, IBM, Microsoft, etc., to handle big data.

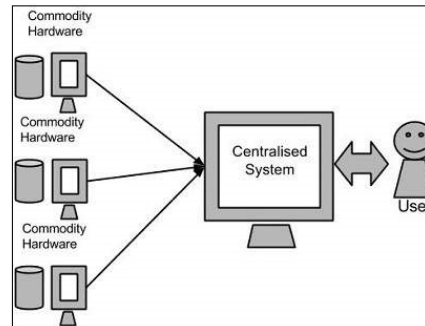
Storage & processing

Traditional Approach



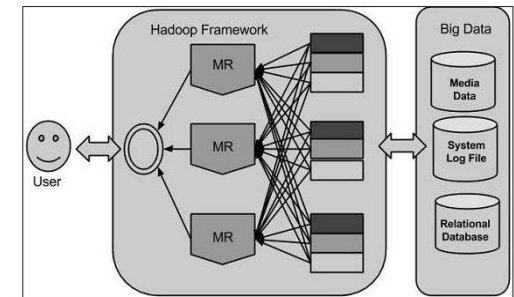
- One computer to store and process
- Select DBMS tool
Oracle, SQL...
- Use interact with the app
- Works on less volume
- Single point of failure
- Not scalable

Google's Solution



- Parallel storage & Processing
Use MapReduce
- divide task into small part
 - Assign each part to d/t PC
 - Collect result from each PC & integrate

Hadoop



- Parallel storage & Processing
Use MapReduce
- divide task into small part
 - Assign each part to more PC
 - Collect result from each PC & integrate



Data Science Tools For Data Storage

Apache Hadoop

- Apache Hadoop is a free, open-source framework that can manage and store tons and tons of data. It provides distributed computing of massive data sets over a cluster of 1000s of computers. It is used for high-level computations
- Here's a list of features of Apache Hadoop:
- Effectively scale large data on thousands of Hadoop clusters
- It uses the Hadoop Distributed File System (HDFS) for data storage which distributes massive amounts of data across several nodes for distributed, parallel computing
- Provides the functionality of other data processing modules, such as Hadoop MapReduce, Hadoop YARN, and s
- and data processing.

Data Science Applications



- Fraud and Risk Detection
- Healthcare
- Internet Search
- Targeted Advertising
- Website Recommendations

- Advanced Image Recognition
- Speech Recognition
- Airline Route Planning
- Gaming
- Augmented Reality



End of Chapter 2



Chapter 3: Artificial Intelligence (AI)



Chapter Outline

 Introduction to AI

 What is AI

- History of AI
- Levels of AI
- Types of AI

 Applications of AI

- Agriculture
- Health
- Business (Emerging market)
- Education

 AI tools and platforms (eg: scratch/object tracking)

 Sample application with hands on activity (simulation based)



What is Artificial Intelligence?

❑ Artificial Intelligence (afterwards call it as **AI**) is composed two words: **Artificial** and **Intelligence**.

AI=Artificial + Intelligence

❑ Artificial defines as “**man-made**” and Intelligence defines the “**thinking power**” or “the ability to learn and solve problems” .

❑ There fore, AI means *"a man-made thinking power."*

❑ So AI defined as: *"It is a branch of Computer Science by which we can create intelligent machines which can behave like **a human, think like humans, and able to make decisions.**"*



Intelligence is composed of :

- ☐ Reasoning

- ☐ Learning

- ☐ Problem Solving

- ☐ Perception

- ☐ Linguistic Intelligence

- AI system composed of **agent** and its **environments**.

- An agent is anything that can:

- ☐ Perceive its environment through sensors.

- ☐ Acts upon that environment through effectors.



- **Intelligent agents** must be able to set goals and achieve them.
- Machine perception is the ability to use input from sensors (**such as cameras, microphones, sensors, etc.**) to deduce aspects of the world. e.g., ***Computer Vision.***



High-profile examples of AI includes:

- ✓ Autonomous vehicles (such as drones and self-driving cars)
- ✓ Medical diagnosis.
- ✓ Creating art (such as poetry).
- ✓ Proving mathematical theorems.
- ✓ Playing games (such as Chess or Go).
- ✓ Search engines (such as Google search).



- ✓ Online assistants (such as Siri, Google Assistance, Alexa, Cortana etc.).
- ✓ Image recognition in photographs.
- ✓ Spam filtering.
- ✓ Prediction of judicial decisions.
- ✓ Targeting online advertisements



Class Exercise : Getting Aid from Google Assistant

- Open Google Assistant from your smart phone
- Next speak the following terms and investigate the result from the **Google Assistant**
 - Call 994
 - Call 0839999565645
 - Open camera
 - Open telegram
 - Open Facebook



Needs of AI

AI needed to:

- ☐ Create expert systems.
- ☐ Helping machines to find solutions for complex problems.

Goal of AI

The main goals of AI are:

- ☐ Replicate human intelligence
- ☐ Solve Knowledge-intensive tasks
- ☐ An intelligent connection of perception and action



- ❑ Building a machine which can perform tasks that requires human intelligence.
- ❑ Creating system which can exhibit intelligent behavior, learn new things by itself, demonstrate, explain, and can advise to its user.

What Comprises to AI?

AI requires the following disciplines :



Mathematics



Biology



Psychology



Sociology



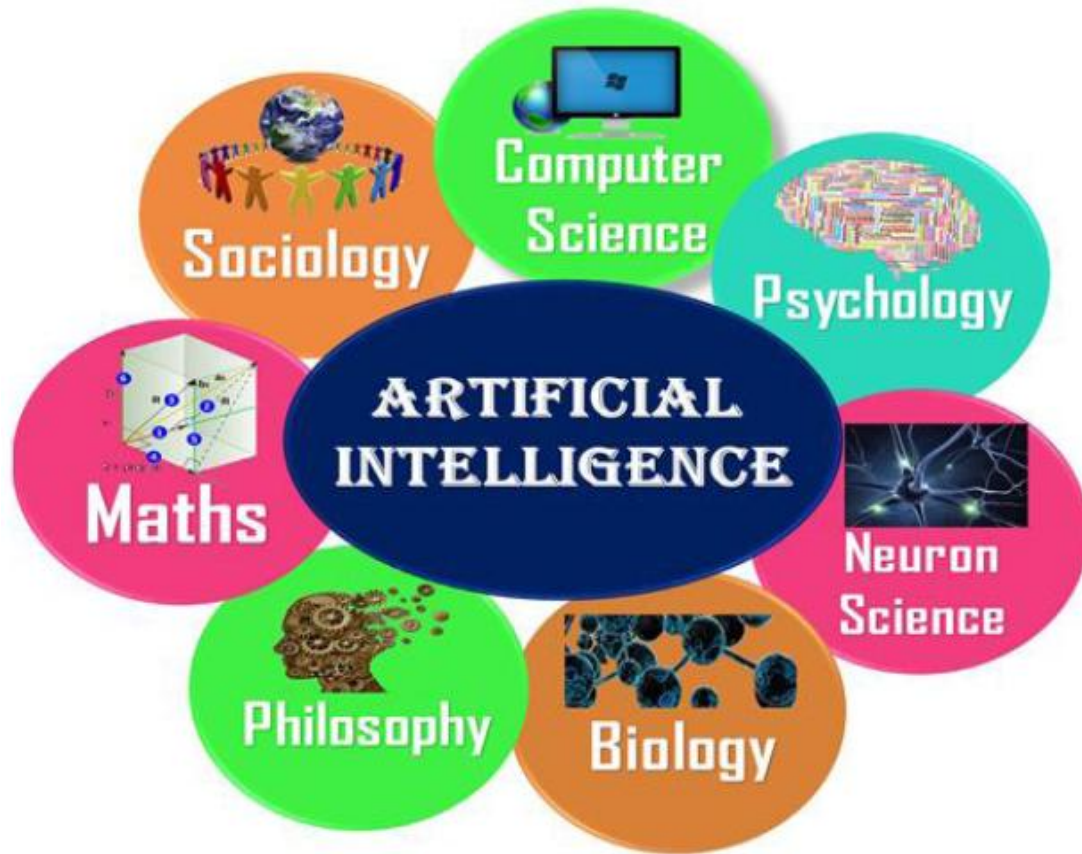
Computer Science



Neurons Study



Statistics





Advantage Vs. Disadvantage of AI

Advantage of AI are:

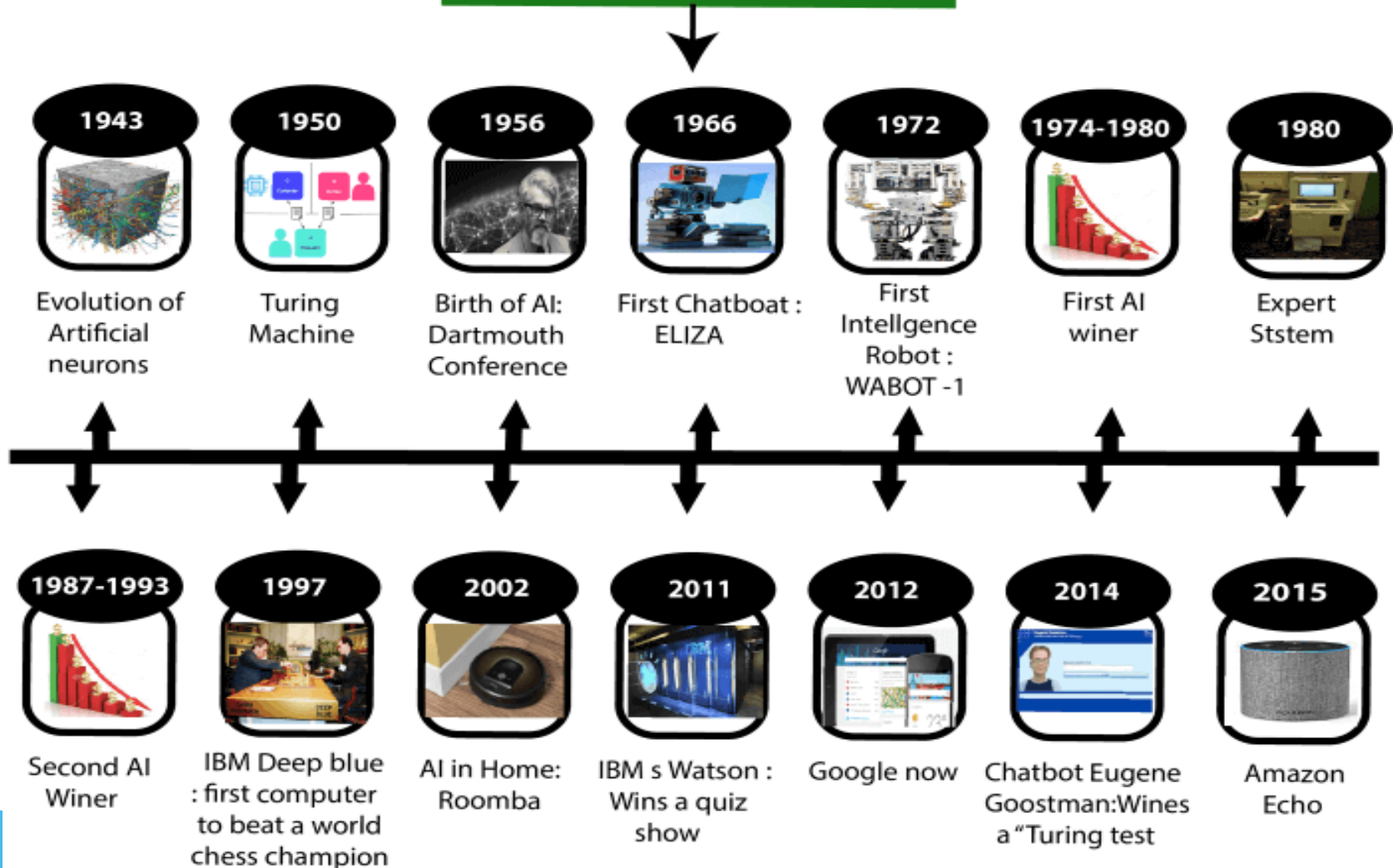
- ☐ High accuracy with fewer errors.
- ☐ High speed.
- ☐ High reliability.
- ☐ Useful for risky areas.
- ☐ Digital Assistant.
- ☐ Useful as a public utility.



Disadvantages of AI:

- ☐ High cost.
- ☐ Can't think out of the box.
- ☐ No feeling and emotions.
- ☐ Increase dependence on machines.
- ☐ No original creativity.

History of AI





Assignment

- *Read about History of AI from page 41-44 of your module and other AI related materials.*
- *Summarize it and prepare a PowerPoint not less than **10** and not more than **12** slides.*
- *Present it to your classmates.*



Levels of AI

Seven levels of AI:

Stage 1 – Rule-Based Systems.

Stage 2– Context Awareness and Retention.

Stage 3 – Domain-Specific Expertise.

Stage 4 – Reasoning Machines.

Stage 5 – Self Aware Systems / Artificial General Intelligence (AGI).

Stage 6 – Artificial Super Intelligence (ASI).

Stage 7 – Singularity and Transcendence.



Stage 1 – Rule-Based Systems

- The most common uses of AI today fit in this bracket.
- Covering everything from business software (Robotic Process Automation) and
- Domestic appliances to aircraft autopilots.



Stage 2– Context Awareness and Retention

- Algorithms that develop information about the specific domain they are being applied in.
- They are trained on the knowledge and experience of the best humans.
- Their knowledge base can be updated as new situations and queries arise.

Example:

- Chatbots and “Roboadvisors”.



Stage 3 – Domain-Specific Expertise

- Going beyond the capability of humans
- These systems build up expertise in a specific context taking in massive volumes of information which they can use for decision making.

Example:

Cancer diagnosis, Google Deepmind's AlphaGo



Stage 4 – Reasoning Machines

- These algorithms have some ability to attribute mental states to themselves and others
- They have a sense of **beliefs, intentions, knowledge, and how their own logic works.**
- They could reason or negotiate with humans and other machines.
- At the moment these algorithms are still in development, however, commercial applications are expected within the next few years.



Stage 5 – Self Aware Systems

- These systems have human-like intelligence
- The most commonly portrayed AI in media
- It is the goal of many working in AI and some believe it could be realized already from 2024.



Stage 6: Artificial Super Intelligence (ASI)

- AI algorithms can outsmart even the most intelligent humans in every domain.
- Logically it is difficult for humans to articulate what the capabilities might be,
- yet we would hope examples would include solving problems we have failed to so far, such as world hunger and dangerous environmental change.



Stage 7 – Singularity and Transcendence

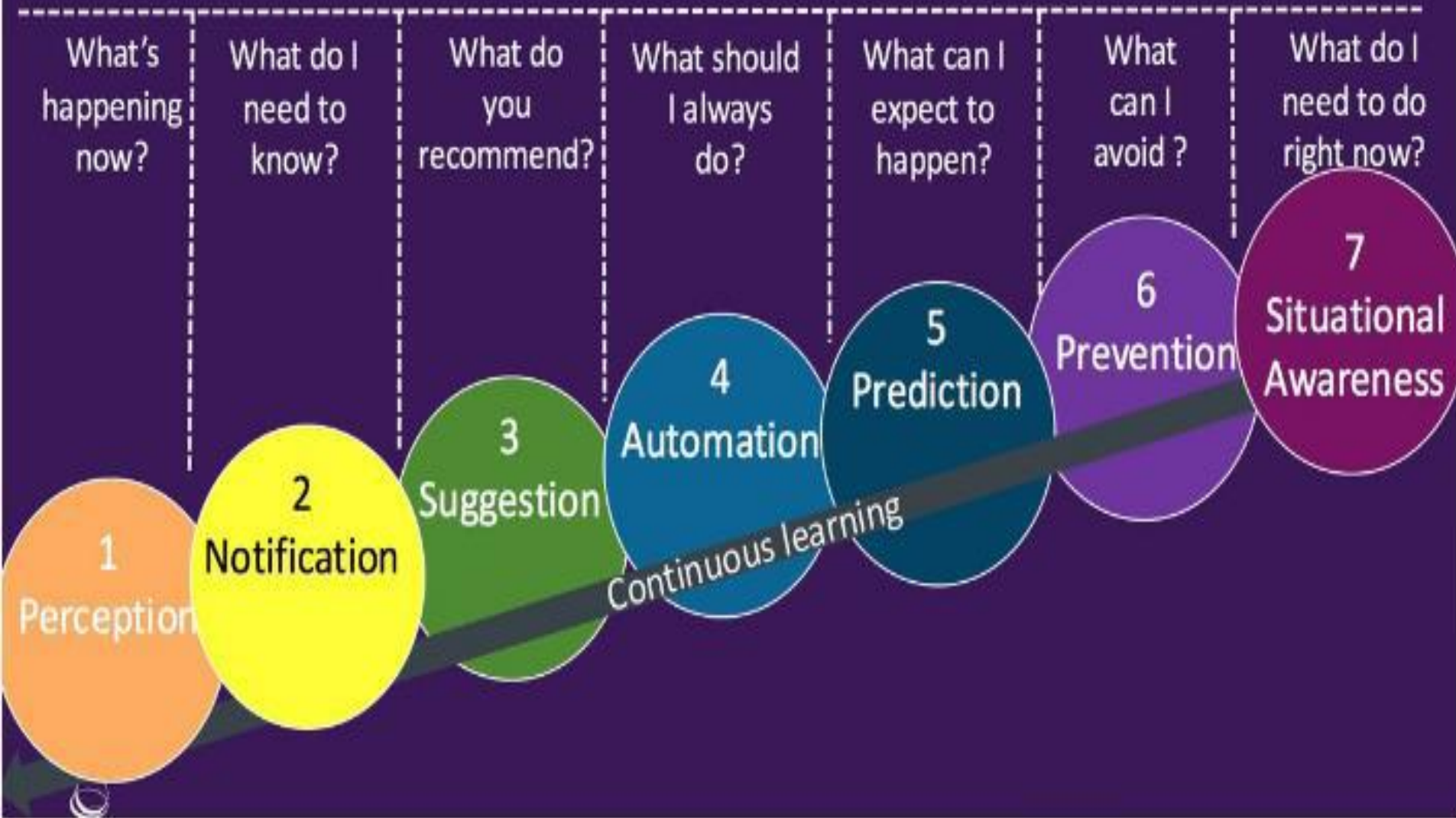
- This is the idea that development provided by ASI (Stage 6) leads to a massive expansion in human capability.
- Human augmentation could connect our brains to each other and to a future successor of the current **internet, creating a “hive mind”** that shares ideas, solves problems collectively, and even gives others access to our dreams as observers or participants. Pushing this idea further, we might go beyond the limits of the human body and connect to other forms of intelligence on the planet



Stage 7 – Singularity and Transcendence

➤ Animals, plants, weather systems, and the natural environment. Some proponents of singularity such as Ray Kurzweil, Google's Director of Engineering, suggest we could see it happen by 2045 as a result of exponential rates of progress across a range of science and technology disciplines. The other side of the fence argues that singularity is impossible and human consciousness could never be digitized.

The seven levels of AI maturity





Type of AI

- AI mainly divide based of **capabilities** and **functions**
- Based on Capabilities AI classified as:
 - ✓ **Narrow(Weak) AI**
 - ✓ **General AI**
 - ✓ **Super (Strong) AI**



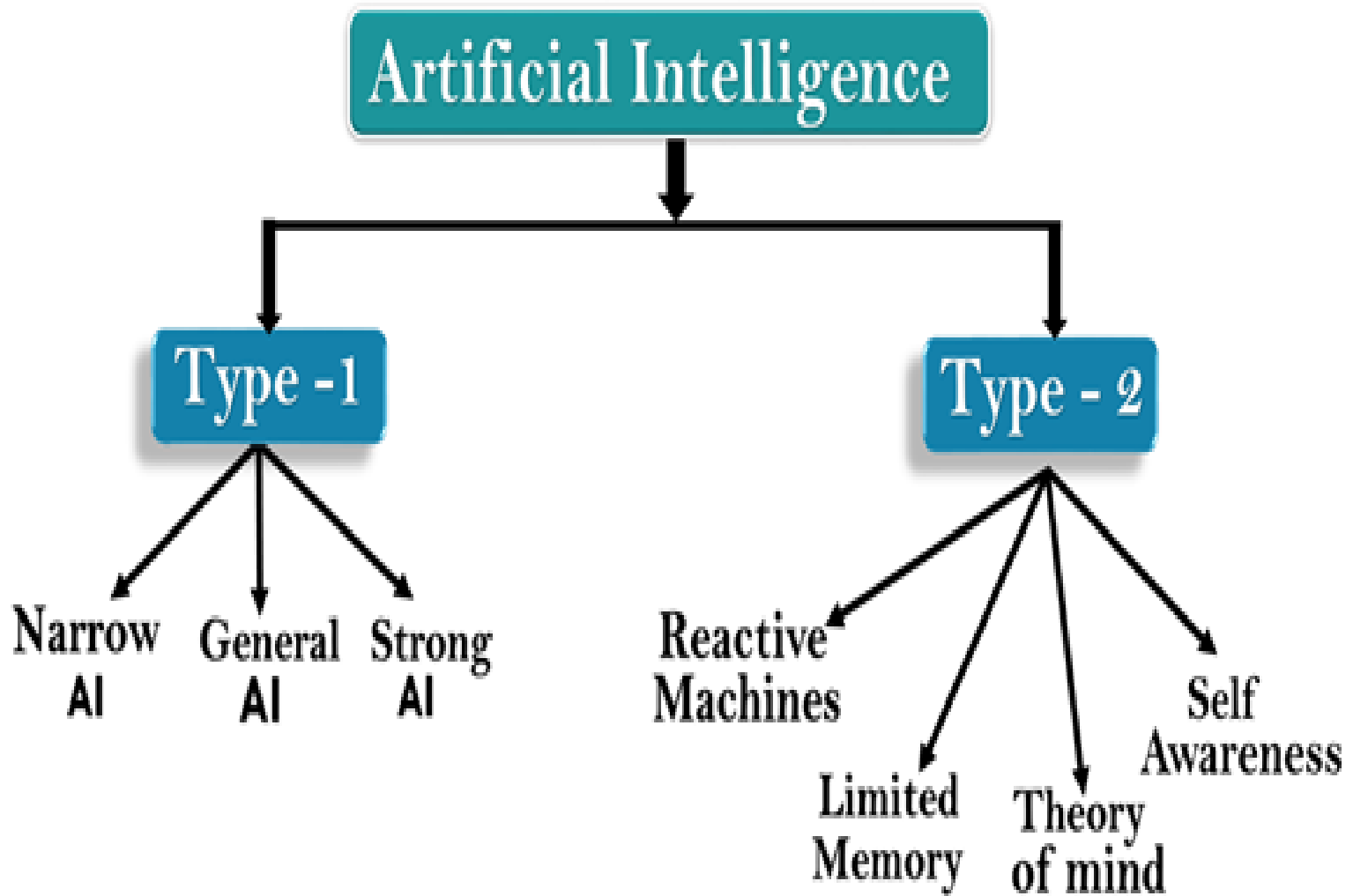
Type of AI

Based on Function AI classified as:

- ✓ **Reactive Machines**
- ✓ **Limited Memory**
- ✓ **Theory of Mind**
- ✓ **Self-Awareness**



Type of AI





How Human Thinks?

- The goal of many researchers is to create **strong** and **general** AI.
- These types of AI learns like a human and can solve general problems as the human brain does.
- To achieve this it requires many years.
- Intelligence or the cognitive process of human is composed of three main stages:
 - ☐ Observe and input the information or data in the brain.
 - ☐ Interpret and evaluate the input.
 - ☐ Make decisions



Mapping human thinking to AI components

- AI is the science of simulating human thinking.
- Because of this it is possible to map human thinking to AI components.

Human Thinking

AI System

Acquire information

Interpreting and evaluating the input data.

Make decision

Sensing layer

Interpretation layer

Interacting layer



Influencers of AI

AI is influenced by :

- ❑ Big data: Structured data versus unstructured data
- ❑ Advancements in computer processing speed and new chip architectures
- ❑ Cloud computing and APIs
- ❑ The emergence of data science



Applications of AI

- AI can solve complex problems in an efficient way in multiple industries
- some sectors which have the application of Artificial Intelligence
 - ✓ AI in agriculture
 - ✓ AI in Healthcare
 - ✓ AI in education:
 - ✓ AI in Finance and E-commerce
 - ✓ AI in Gaming
 - ✓ AI in Social Media
 - ✓ AI in Data Security
 - ✓
 - ✓ AI in Travel &Transport
 - ✓ AI in the Automotive Industry
 - ✓ AI in Robotics:
 - ✓ AI in Entertainment



AI tools and platforms

- AI platforms are defined as some sort of hardware architecture or software framework , that allows the software to run.
- Artificial intelligence (AI) platforms provide users a tool kit to build intelligent applications
- Some platforms offer pre-built algorithms and simplistic workflows, others require a greater knowledge of development and coding



Functionalities of AI Algorithms

Some functionality of the algorithms are:

- Image Processing
- Natural language processing (information retrieval, text mining, question answering, machine translation ...etc)
- Voice recognition
- Recommendation system
- Predictive analytics



AI tools:

- ✓ Search and optimization
- ✓ Logic
- ✓ Probabilistic methods for uncertain reasoning
- ✓ Classifiers and statistical learning methods
- ✓ Neural networks
- ✓ Control theory
- ✓ Languages



Simple AI application

1. Commuting

- Google's AI-Powered Predictions
- Ridesharing Apps Like Uber and Lyft
- Commercial Flights Use an AI Autopilot

2. Email

- Spam Filters
- Smart Email Categorization

3. Social Networking

4. Online Shopping

5. Mobile Use

- Voice-to-Text
- Smart Personal Assistants(Siri and Google Now)
- Alexa, an AI-powered personal assistant that accepts voice commands to create to-do lists, order items online, set reminders, and answer questions



End of Chapter 3



Chapter 4: Internets of Things (IoT)





Content Outline:

4.1. Overview of IoT

4.2. How does it work?

4.3. IoT Tools and Platforms



Chapter Objectives:

➤ **After accomplishing this chapter, Students will be able to:**

- Describe IoT
- Explain the history of IoT
- Describe the pros and cons of IoT
- Explain how IoT works
- Explain the architecture of IoT
- Describe IoT tools and platforms
- Describe some of the application areas of IoT



4.1. Overview of IoT

➤ **IoT features includes:-**

Artificial Intelligence

- Makes virtually anything “smart”.

Connectivity:

- IoT creates small but practical networks that no longer tied to major providers between its system devices.

Sensors :

- IoT loses its distinction without sensors
- makes IoT an active system capable of real-world integration



Active engagement

- IoT introduces a new paradigm for active content, product, or service engagement.

small device use

- IoT exploits purpose-built small devices to deliver its precision, scalability, and versatility



4.1.1. What is IoT?

1) IoT is the networking of smart objects

----- Internet Architecture Board's (IAB)

2) IoT is the networking of smart objects in which smart objects have some constraints such as limited bandwidth, power, and processing accessibility for achieving interoperability among smart objects.

----- Internet Engineering Task Force (IETF)



Cont'd...

3) IoT is a framework of all things that have a representation in the presence of the internet in such a way that new applications and services enable the interaction in the physical and virtual world in the form of Machine-to-Machine (M2M) communication in the cloud.

----Institute of Electrical and Electronics Engineers (IEEE)



- 4) IoT is the interaction of everyday object's computing devices through the Internet that enables the sending and receiving of useful data.

----Oxford dictionary's definition

- 4) IoT is expressed through a simple formula such as:

IoT = Services + Data + Networks + Sensors

-----IoT in 2020



- Generally IoT is the network of physical objects or "things" embedded with electronics, software, sensors, and network connectivity, which enables objects to collect and exchange data.
- It is a network of devices that can sense, accumulate and transfer data over the internet without any human intervention.
- It consists of any device with an on/off switch connected to the Internet
- Applications: smart-city, smart-home, smart-energy, connected car, smart agriculture, health care, logistics, connected campus, building & industry etc.

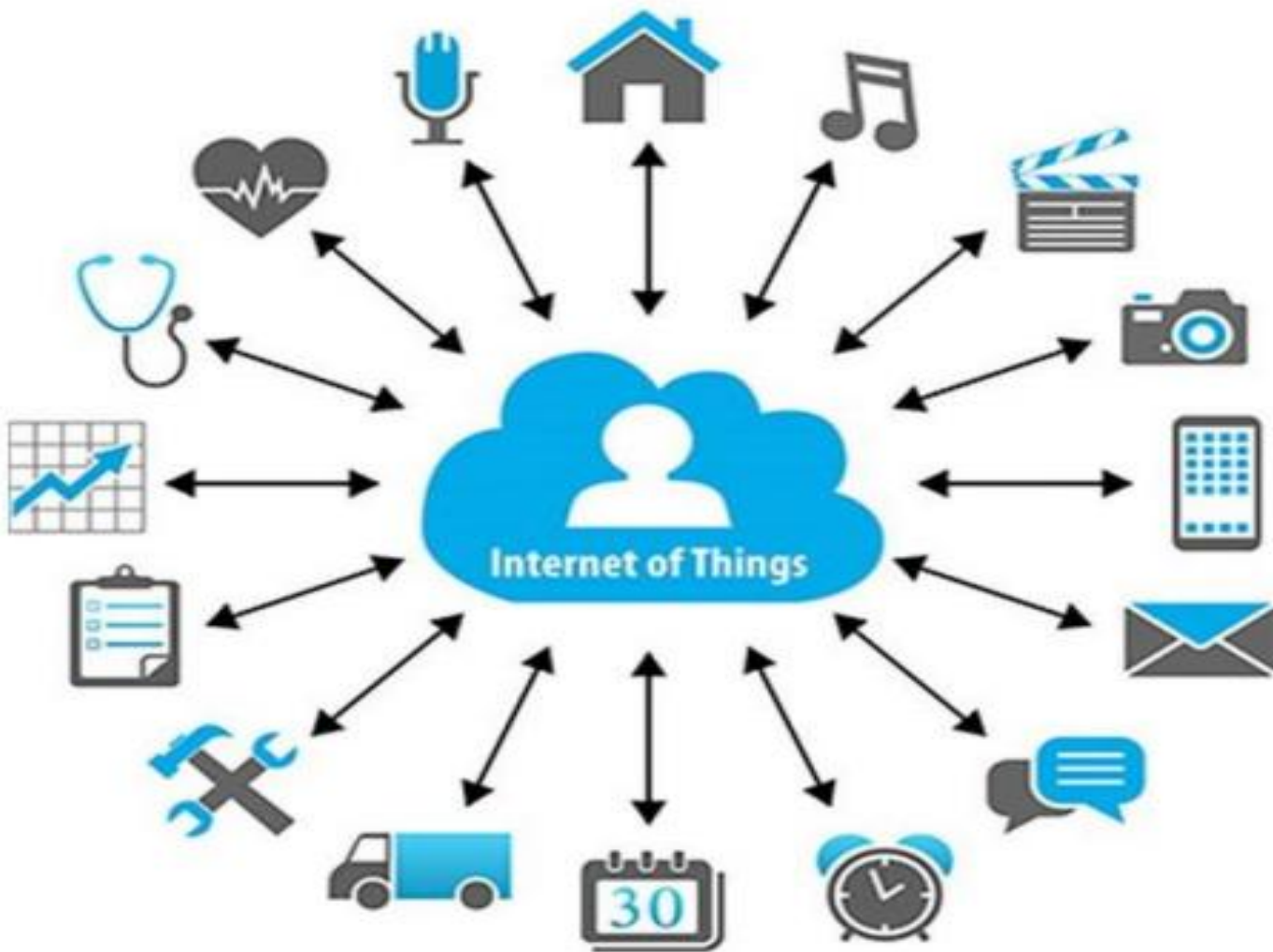


Figure 4.1:- IoT domains



4.1.3. IoT – Advantages

- Improved Customer Engagement
- Technology Optimization
- Reduced Waste
- Enhanced Data Collection

4.1.4. IoT – Disadvantages

- Lack of Security.
- A bug in the system, could affect all connected device



- No international standard of compatibility for IoT,
- Heterogeneous devices couldn't make communication.
- Collecting and managing the data from millions of IoT devices will be challenging for a given Enterprise.

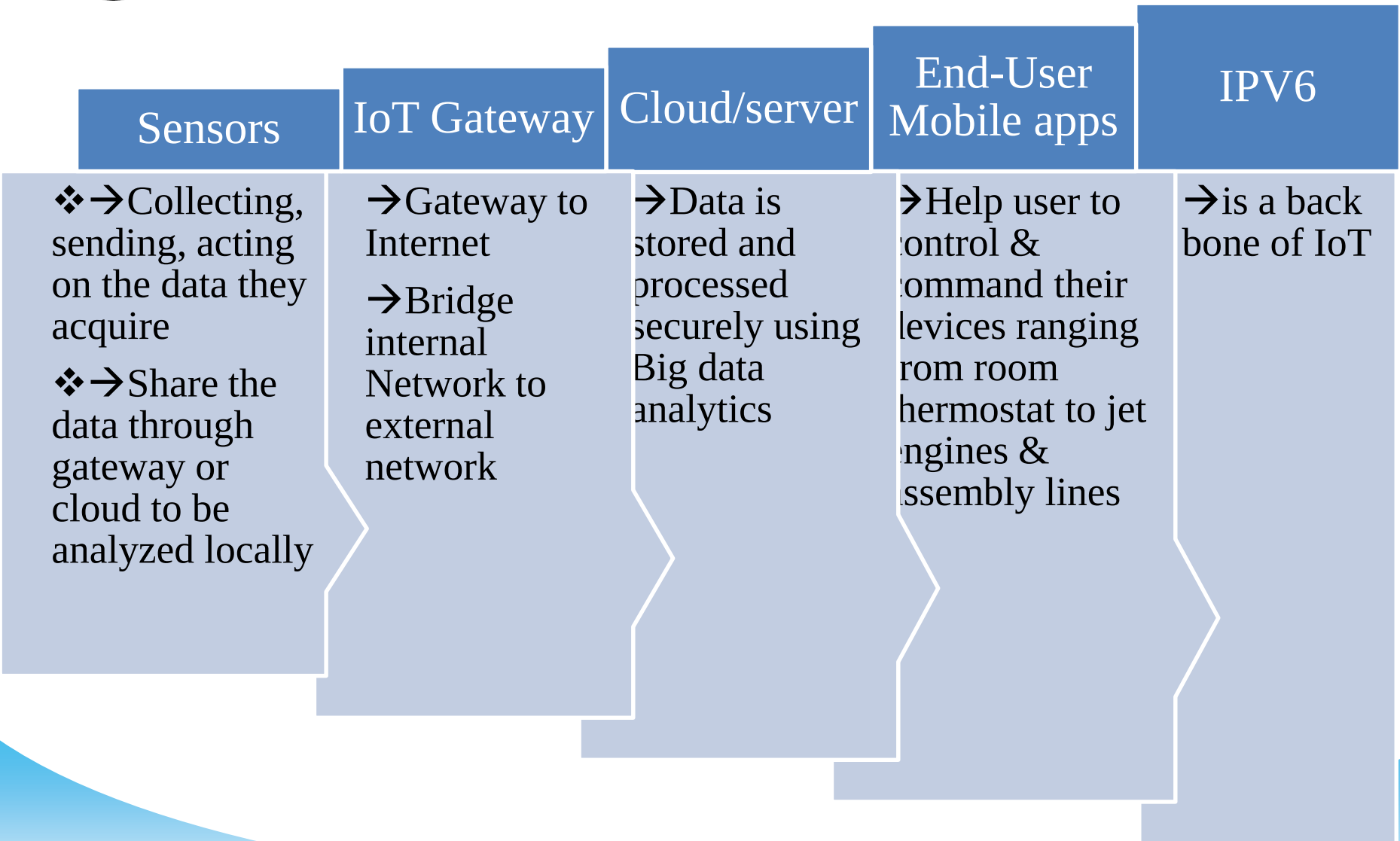


4.1.5. Challenges of IoT

- ☐ Security
- ☐ Privacy
- ☐ Complexity
- ☐ Flexibility
- ☐ Compliance

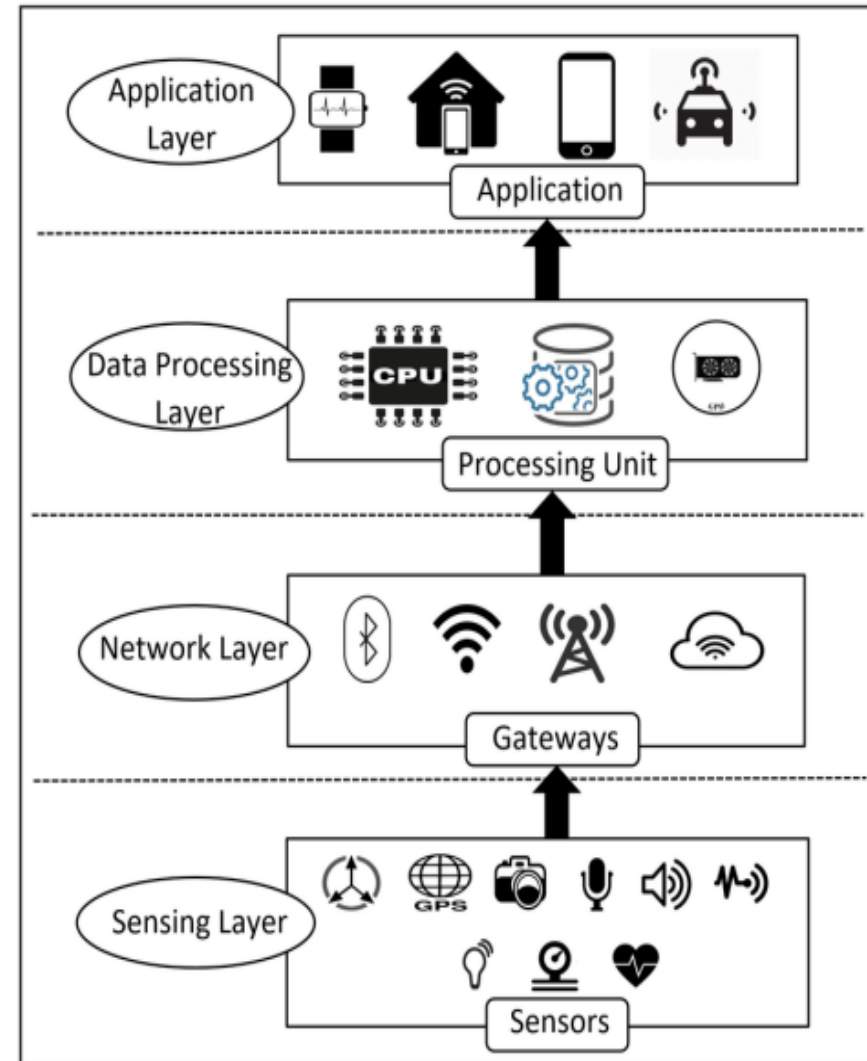


4.2. How does it work?



4.2.1. Architecture of IoT

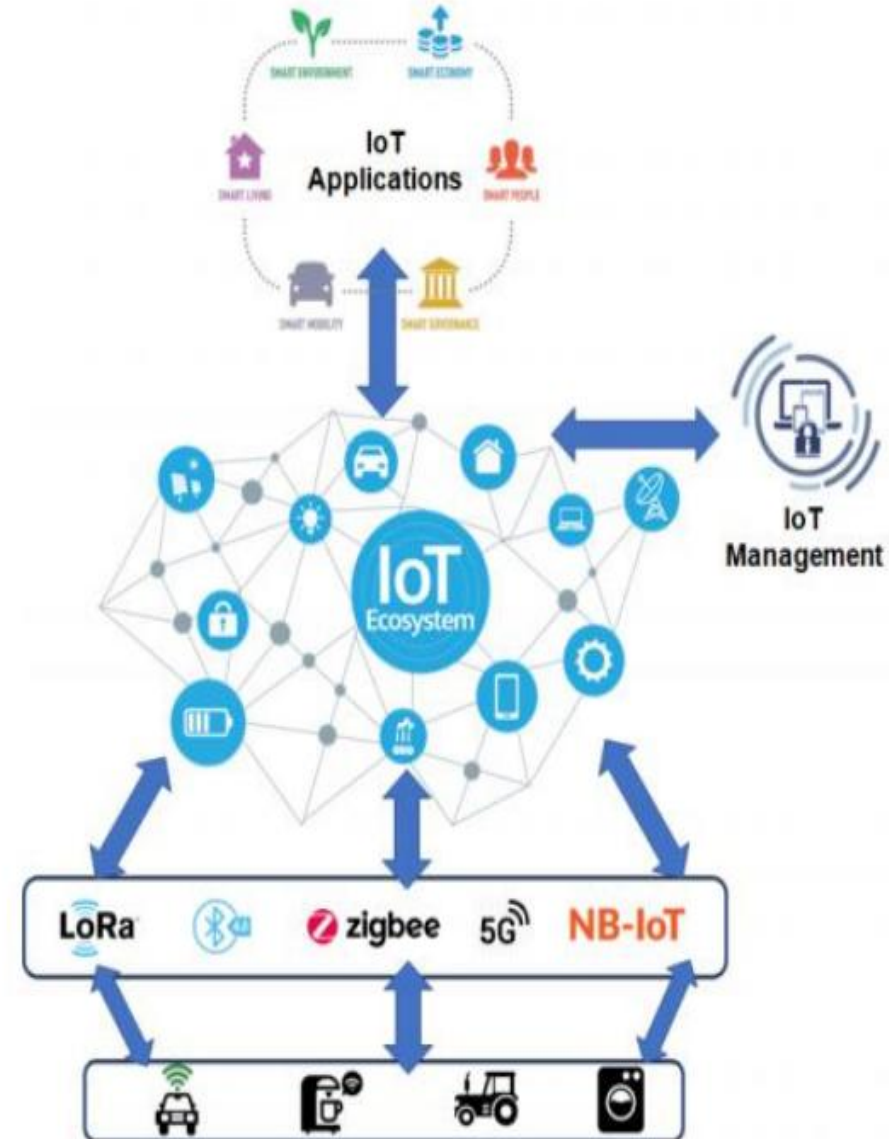
- ❑ Sensing Layer
- ❑ Network Layer
- ❑ Data Processing Layer
- ❑ Application Layer



4.2.2. Devices and Networks

➤ Devices can be categorized into three main groups:

- 1) Consumer
- 2) Enterprise
- 3) Industrial





Cont'd...

- IoT network devices has constrained resources (power, processing, memory, etc.)
- The IoT network devices must be managed, configured and monitored remotely to ensure their correct functioning.
- IoT network and device management solution should consider several elements



4.3. IoT Tools and Platforms

Key Features:

- Manage an unlimited number of connected devices
- Set up cross-device interoperability
- Perform real-time device monitoring
- Perform remote device provisioning & configuration
- Collect and analyze sensor data
- Analyze user behavior and deliver targeted notifications
- Create cloud services for smart products

KAA



Site Where

Key Features:

- Run any number of IoT applications on a single SiteWhere instance
- Spring delivers the core configuration framework
- Add devices through self-registration
- Integrates with third-party integration frameworks such as Mule any point
- Default database storage is MongoDB
- Eclipse Californium for CoAP messaging
- InfluxDB for event data storage
- Grafana to visualize SiteWhere data



Thing Speak

Key Features:

- Collect data in private channels
- Share data with public channels
- MATLAB analytics and visualizations
- Alerts
- Event scheduling
- App integrations
- Worldwide community



Device Hive

Key Features:

- Directly integrate with Alexa
- Visualization dashboard of your choice
- It supports Big data solutions such as Elastic Search, Apache Spark, Cassandra and Kafka for real-time and batch processing.
- Connect any device
- It comes with Apache Spark and Spark Streaming support.
- Supports libraries written in various programming languages, including Android and iOS libraries
- It allows running batch analytics and machine learning on top of your device data



Zetta

Key Features:

- Supports a wide range of hacker boards
- Zetta allows you to assemble smartphone apps, device apps, and cloud apps

Things Board

Key Features:

- Real-time data visualization and remote device control
- Customizable rules, plugins, widgets and transport implementations 76
- Allows monitoring client-side and provision server-side device attributes.
- Support multi-tenant installations out-of-the-box.
- Supports transport encryption for both MQTT & HTTP(s) protocols.



4.4. Applications of IoT

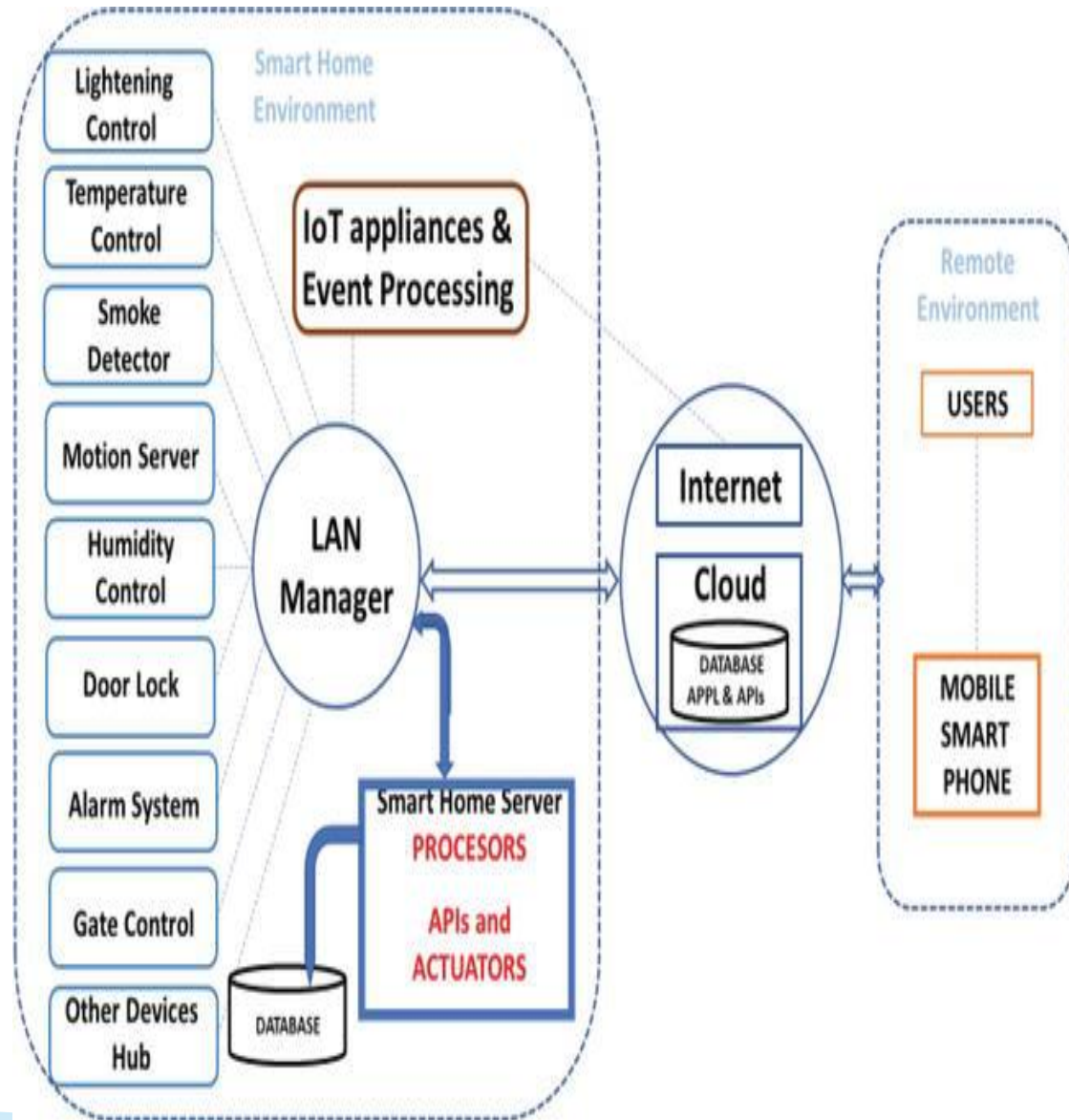
IoT is applicable in sectors :

- ☐ Agriculture
- ☐ Consumer Use
- ☐ Healthcare
- ☐ Insurance
- ☐ Manufacturing
- ☐ Retail
- ☐ Transportation
- ☐ Utilities



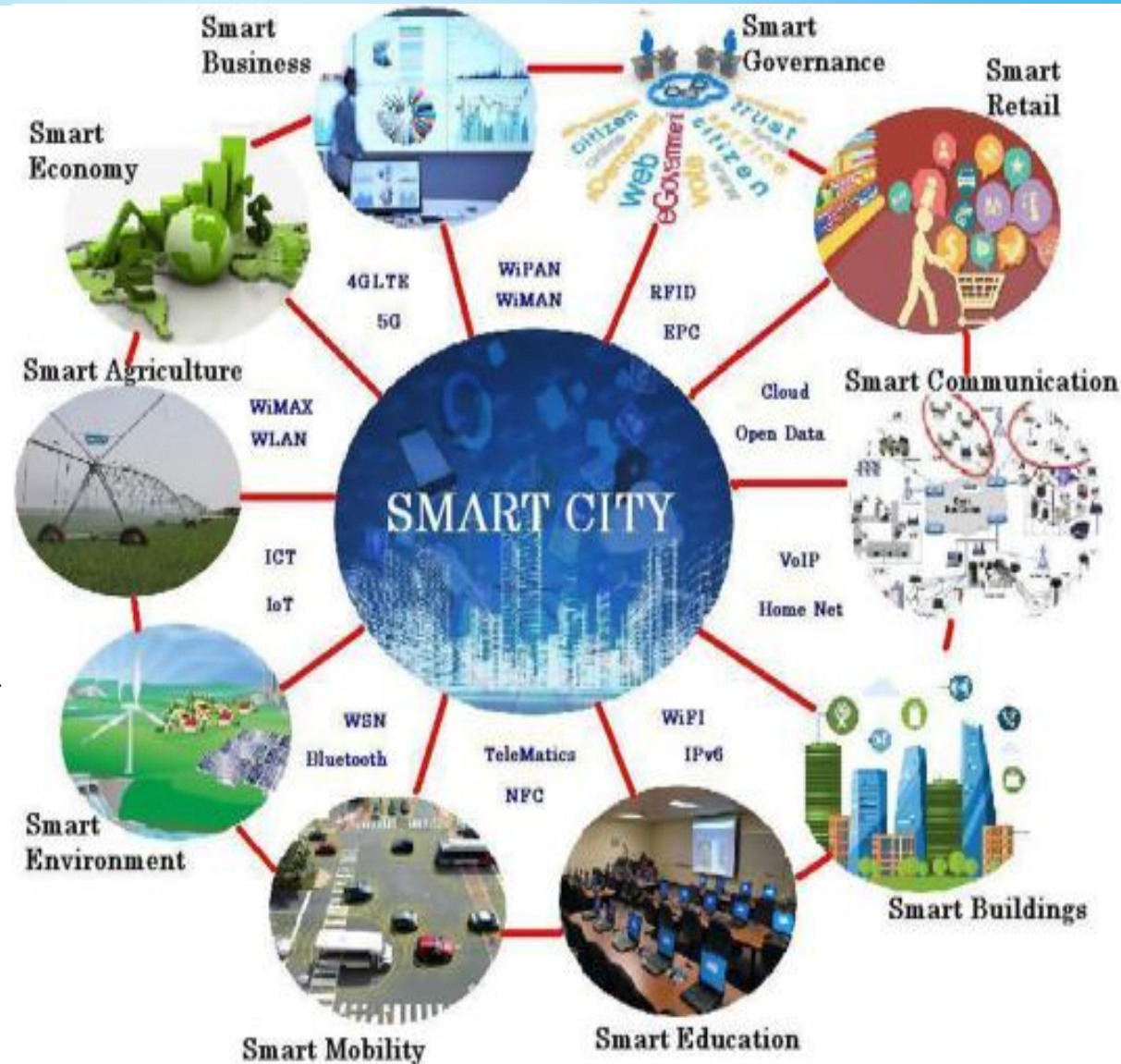
4.3.1. IoT Based Smart Home

- ☐ Remote Control Appliances
- ☐ Weather Controlling
- ☐ **Smart Home Appliances**
- ☐ Energy and Water Use
- ☐ Intrusion Detection Systems
- ☐ Safety Monitoring



4.3.2. IoT Based Smart City

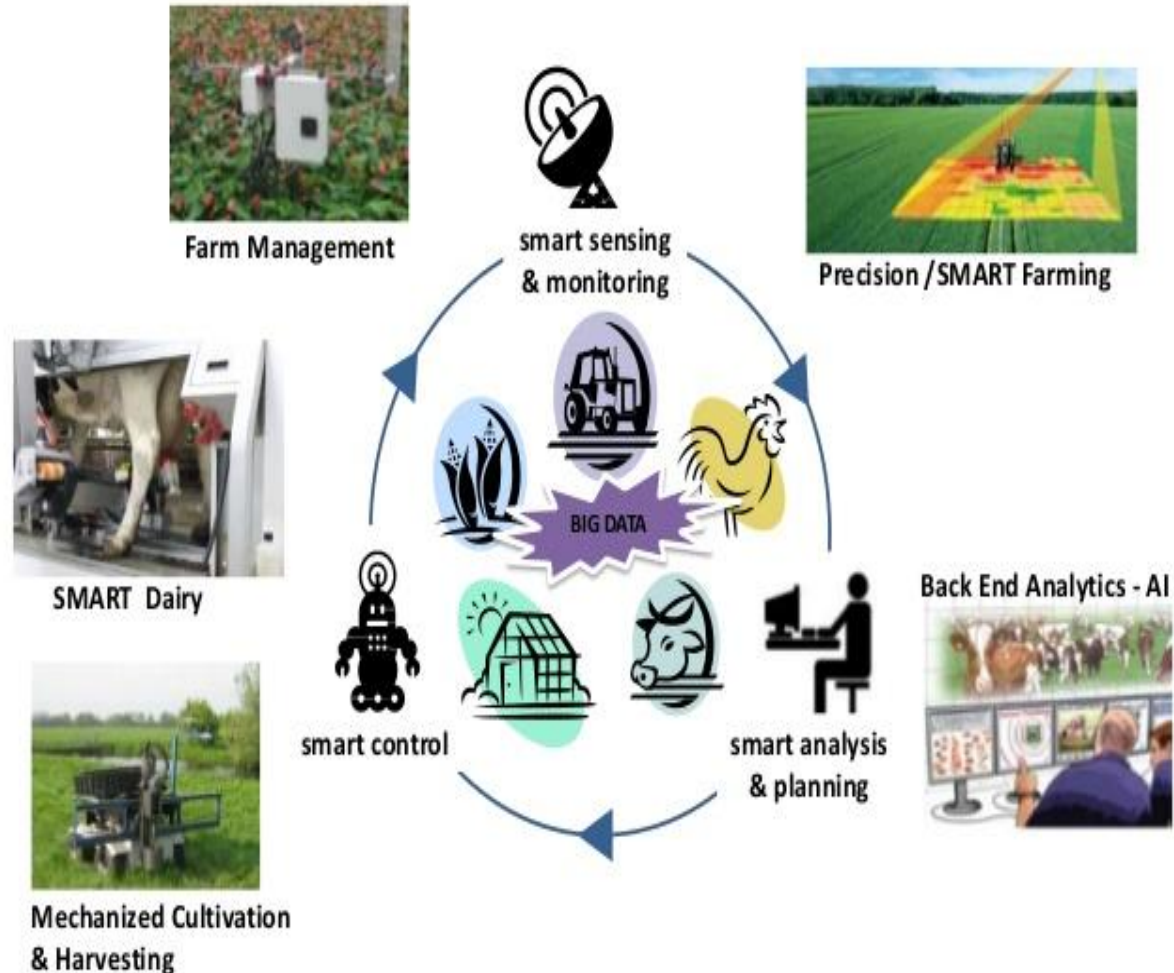
- Structural Health
- Lightning
- Safety
- Transportation
- Waste Management
- Smart Parking



4.3.3. IoT Based Smart Farming

- ☐ Green Houses
- ☐ Compost
- ☐ Animal Farming/Tracking
- ☐ Offspring Care
- ☐ Field Monitoring

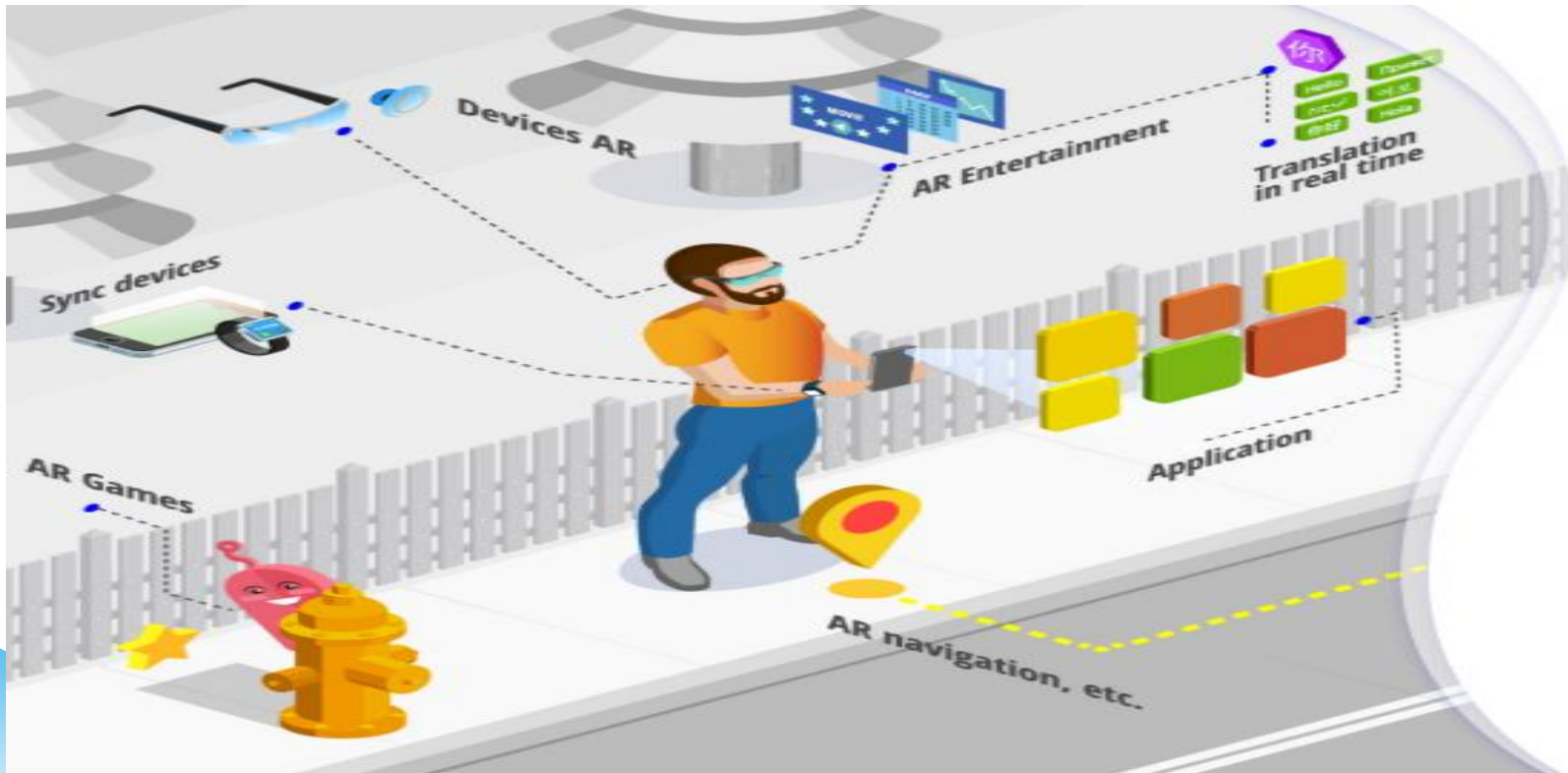
Future SMART Agriculture using IoT – Big PictureView



With Acknowledgements due to WUR University



Chapter 5: Augmented Reality(AR)





Content outlines:

- ☐ Overview of Augmented Reality.
- ☐ The difference between VR,AR and MR
- ☐ Architecture of AR systems
- ☐ Application of AR systems



Chapter Objectives:

- After accomplishing this chapter, Students will be able to:
 - ☐ Explain augmented reality
 - ☐ Explain the features of augmented reality
 - ☐ Explain the difference between AR, VR, and MR
 - ☐ Explain the architecture of augmented reality systems
 - ☐ Describe the application areas of augmented reality



Brainstorming

- What do you think about Augmented Reality?
- What are the common features of Augmented reality?



- The fundamental idea of AR is to combine , or mix, the view of the real environment with additional, virtual content that is presented through computer graphics.
- Its convincing effect is achieved by ensuring that the virtual content is aligned and registered with the real objects.
- As a person moves in an environment and their perspective view of real objects changes, the virtual content should also be presented from the same perspective.



What is Augmented reality(AR)?

Definition

- **Augmented reality (AR)** is a form of emerging technology that allows users to overlay computer generated content in the real world.
- AR refers to a live view of a physical real-world environment whose elements are merged with augmented computer-generated images creating a mixed reality.



- **Augmented Reality (AR)** as a real-time direct or indirect view of a physical real-world environment that has been enhanced /augmented by adding virtual computer-generated information to it.
- Augmented reality is the integration of digital information with the user's environment in real time.



Virtual Reality (VR)

- **Virtual Reality (VR)** is fully immersive, which tricks your senses into thinking you're in a different environment or world apart from the real world.
- Using a head-mounted display (HMD) or headset, you'll experience a computer-generated world of imagery and sounds. you can manipulate objects and move around using haptic controllers while tethered to a console or PC.
- It is also called a computer-simulated reality.



- It refers to computer technologies using reality headsets to generate realistic sounds, images and other sensations that replicate a real environment or create an imaginary world.
- Advanced VR environment will engage all five senses (taste, sight, smell, touch, sound) .
- Using VR devices such as HTC Vive, Oculus Rift or Google Cardboard, users can be transported into a number of real-world and imagined environments.



Figure 5.3 VR Case that Inserts a Smartphone



- Most VR headsets are connected to a computer (Oculus Rift) or a gaming console (PlayStation VR)
- There are standalone devices (Google Cardboard is among the most popular) as well.
- Most standalone VR headsets work in combination with smartphones – you insert a smartphone, wear a headset, and immerse in the virtual reality



Augmented Reality (AR)

- In augmented reality, users see and interact with the real world while digital content is added to it.
- There's a different way to experience augmented reality, though – with special AR headsets, such as
- Google Glass, where digital content is displayed on a tiny screen in front of a user's eye.
- AR adds digital elements to a live view often by using the camera on a smartphone.
- Examples of augmented reality experiences include Snapchat lenses and the game Pokémon Go.



- Augmented Reality (AR) is a live, direct or indirect view of a physical, real-world environment whose elements are augmented (or supplemented) by computer-generated sensory input such as sound, video, graphics or GPS data.



Mixed Reality (MR)

- **Mixed Reality (MR)** sometimes referred to as hybrid reality.
- It is the merging of real and virtual worlds to produce new environments and visualizations where physical and digital objects co-exist and interact in real-time.
- For example, with MR, you can play a virtual video game, grab your real-world water bottle, and smack an imaginary character from the game with the bottle.
- Imagination and reality have never been so intermingled.

- In mixed reality, you interact with and manipulate both physical and virtual items and environments, using next-generation sensing and imaging technologies.



- MR allows you to see and immerse yourself in the world around you even as you interact with a virtual environment using your own hands.





VR Vs. AR Vs. MR

- The hardware requirements VR is content which is 100% digital and can be enjoyed in a fully immersive environment
- AR overlays digital content on top of the real-world.
- MR is a digital overlay that allows interactive virtual elements to integrate and interact with the real-world environment.

AR vs VR vs MR

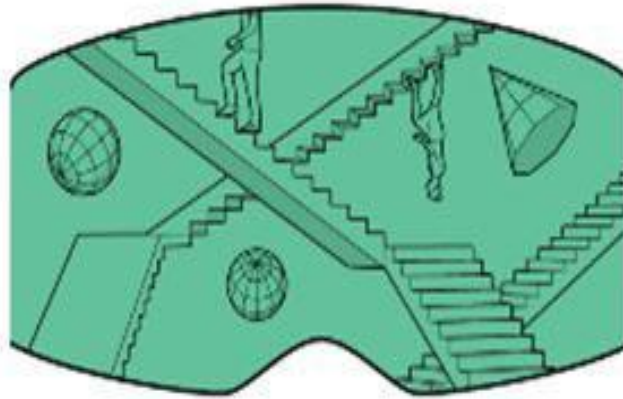
Characteristics	VR	AR	MR
Is user aware about real world ?	NO	YES	YES
Can user interact with Real and Virtual world in real-time ?	NO	YES	YES
Can Real and Virtual contents interact with each other in real time ?	NO	NO	YES



- Numerous augmented reality apps and games can run on almost every smartphone on the market.
- virtual reality programs require specialized VR headsets, noise-canceling headphones, cameras to track room space and boundaries, and sometimes even motion capture technology.
- Some of the biggest names in VR tech today are the Oculus Rift, HTC Vive, and PlayStation VR.
- For the enjoyment of simple VR videos, there are affordable makeshift VR headsets like the Google Cardboard.



- Mixed reality hardware is still emerging and hasn't quite broken into the mainstream consumer market, most likely due to the price.
- The consumer releases of the Microsoft HoloLens and Magic Leap One retail for over \$2000 USD .
- However, mixed reality applications sometimes require exponentially more processing power and thus require more powerful hardware.



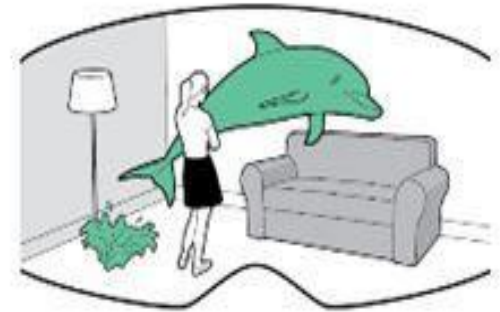
Virtual Reality

VR places the user in another location entirely. Whether that location is computer-generated or captured by video, it entirely occludes the user's natural surroundings.



Augmented Reality

In augmented reality—like Google Glass or the Yelp app's Monocle feature on mobile devices—the visible natural world is overlaid with a layer of digital content.



Mixed Reality

In technologies like Magic Leap's, virtual objects are integrated into—and responsive to—the natural world. A virtual ball under your desk, for example, would be blocked from view unless you bent down to look at it.

VR vs. AR vs. MR



VR

*Anything that
is not real*

VIRTUAL REALITY



AR

*Augmenting virtual objects in
the real world*

AUGMENTED REALITY



MR

*It is a hybrid reality -
a mix of AR and VR*

MIXED REALITY



➤ The Microsoft HoloLens includes:

- ☐ A built-in microphone array,
- ☐ Binaural sound capabilities,
- ☐ A built-in camera for recording,
- ☐ A depth sensor,
- ☐ Head-tracking cameras,
- ☐ An inertial measurement unit which helps track head movement



The architecture of AR Systems

- The first Augmented Reality Systems (ARS) were usually designed with a basis on three main blocks
1. Infrastructure Tracker Unit,
 2. Processing Unit, and
 3. Visual Unit.



The Infrastructure Tracker Unit

- It is responsible for collecting data from the real world, sending them to the Processing Unit,

Processing Unit

- Which mixed the virtual content with the real content and sent the result to the Video Out module of the Visual Unit

Visual Unit

- Used to display

➤ Some designs used a Video In, to acquire required data for the Infrastructure Tracker Unit.

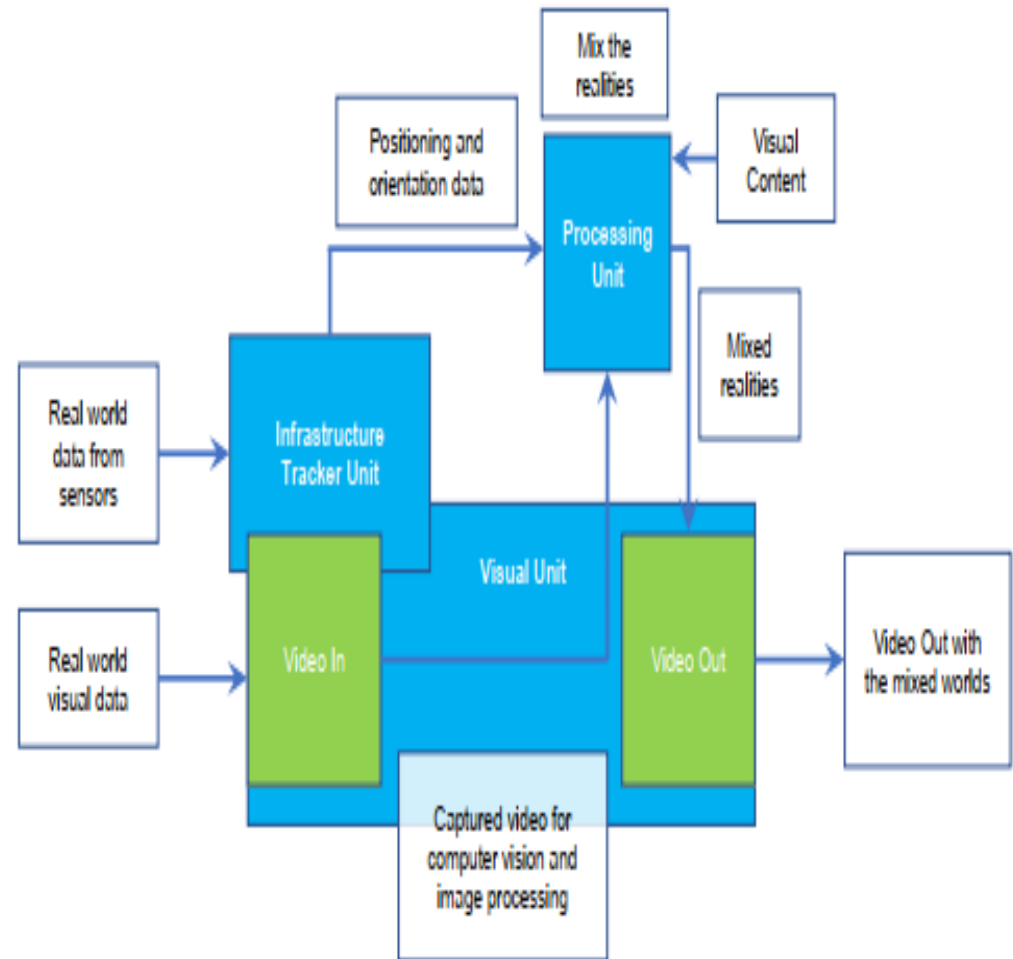


Figure 5.7 Augmented Reality Systems (ARS) standard architecture



- The Visual Unit can be classified into two types of system, depending on the followed visualization technology:

1. **Video see-through**

- ☐ It uses a Head-Mounted Display (HMD) that employs a video-mixing and displays the merged images on a closed-view HMD.

2. **Optical see-through:**

- ☐ It uses an HMD that employs optical combiners to merge the images within an open-view HMD.



HMDs shortcomings

- HMDs are currently the dominant display technology in the AR field.
- However, they lack in several aspects, such as ergonomics, high prices and relatively low mobility due to their sizes and connectivity features.
- An additional problem involving HMD is the interaction with the real environment, which places virtual interactive zones to the user, making the collision with these zones hard due to the difficulty to interact with multiple points in different depths.



Applications of AR Systems

- Technology is ever-changing and ever-growing.
- One of the newest developing technologies is augmented reality (AR), which can be applied to many different disciplines such as
 - Education.
 - Medicine.
 - Entertainment.
 - Military, etc.



AR In education

- Augmented reality allows flexibility in use that is attractive to education.
- AR technology can be utilized through a variety of mediums including desktops, mobile devices, and smartphones.
- AR can be used to enhance content and instruction within the traditional classroom,
- supplement instruction in the special education classroom, extend content into the world outside the classroom,



➤ More importantly, the following reasons for using augmented reality in education:

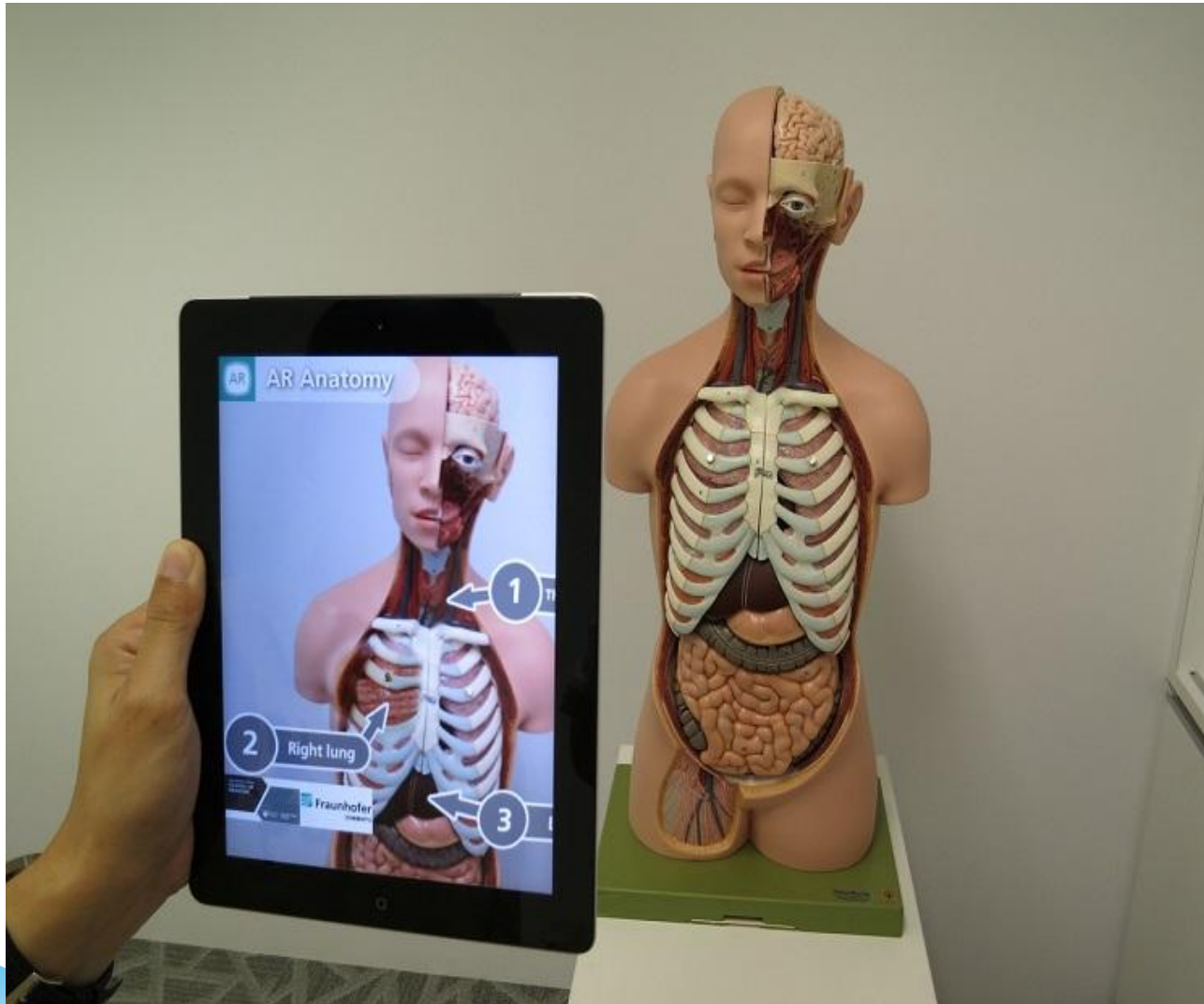
- ☐ Affordable learning materials
- ☐ Interactive lessons
- ☐ Higher engagement
- ☐ Higher retention
- ☐ Boost intellectual curiosity



AR In Medicine

- Augmented reality has the potential to play a big role in improving the healthcare industry.
- This new technology is enhancing medicine and healthcare towards more safety and efficiency.
- For now, augmented reality has already made significant changes in the following medical areas:
 - ☐ Surgery(minimally invasive surgery);
 - ☐ Education of future doctors;
 - ☐ Diagnostics;

AR in health care





➤ AR tools aid to detect the signs of depression and other mental illnesses by reading from

- ☐ Facial expressions.

- ☐ Voice tones.

- ☐ Physical gestures



➤ In medicine, AR has the following applications:

☐ Describing symptoms

☐ Nursing care

☐ Surgery

☐ Ultrasounds

☐ Diabetes management

☐ Navigation



- Generally, AR provides the following benefits to patients and healthcare workers:
- ❑ Reduce the risks associated with minimally invasive surgery.
 - ❑ Better informed decisions about the right treatment and illness prevention.
 - ❑ Make procedures more tolerable.
 - ❑ Better aftercare
 - ❑ Medical training and education.
 - ❑ Assistance in medical procedures and routine tasks.



- AR in games :- such as Pokémon Go
- AR in music
- AR on TV
- AR in eSports
- AR in the theater



Review Questions

1. what is augmented, virtual and mixed reality and its application?
2. what are good examples of augmented, virtual and mixed reality?
3. what is the difference between augmented, virtual and mixed reality?
4. How is augmented, virtual and mixed reality achieved?
5. What is the benefit of augmented, virtual and mixed reality?



6. How Can AR, VR, and MR improve engineering instructions?
7. Can VR be a substitute for Real Life Experience?
8. What is the impact of VR on Educational Learning rather than games?
9. What is the most technical challenge for MR?



Objectives of the chapters:

- Distinguish the link between ethics and technology.
- Understand general, professional and leadership ethical questions.
- Explain what digital privacy is, its components and why it is important.
- Know the importance of accountability and trust in emerging technologies.
- Identify the threats and challenges we face in developing and utilizing emerging technologies.



6.2. New ethical questions

6.1. Technology and ethics

Activity 6.1

- From your civic and ethical education course, what do you understand about the word ethics?



6.1 Technology and ethics

- ❖ Importance of Internet as a technology:
 - Provide new tools and new ways for people to interact.
- ❖ However brought a drawback Like:
 - New concerns about privacy,
 - spam and viruses,
 - Digital divide across the universe.
- ❖ This takes us to the point that technology can promote or restrict the human right.



6.1 Technology and ethics

- ❑ For the Information Society, the concerns lies on how to maximize the benefits while minimizing the harms in using ET.
- The compelling solution is to set up a proper legal or regulatory system for enhanced gain of benefits for the larger mass, that generally takes us to Ethical values.



6.1 Technology and ethics

- ❖ Ethics has become a crucial aspect of professional accountability with the inclusion of five basic principles
 - ☐ Integrity
 - ☐ Objectivity
 - ☐ Competence and due care.
 - ☐ Confidentiality, and
 - ☐ Professional behavior.
- ❖ However, the emergence of new technologies raises some new challenges for the profession to address.



6.2. New ethical questions

6.2. New ethical questions

Activity 6.2

What do you think the need for ethics in data science? Is it really important to include ethical rules when dealing with big data? If your answer is yes, why?



❖ The increasing use of

- ❑ Big data.
- ❑ Algorithmic decision-making, and
- ❑ Artificial intelligence can enable more
 - consistent. ,
 - evidence-based and
 - accurate judgments or decisions.

❖ Nevertheless the concern still remains how we could possibly and ethically use the emerging new technologies.



❖ The concern here is:- If we deal with biased data, or data based on personal preferences and activities, we would come up with biased and discriminatory result that might intern lead to unfair and unethical out come.

Activity 6.3

What do you think the need for ethics in data science? Is it really important to include ethical rules when dealing with big data? If your answer is yes, why?



6.2. New ethical questions

❖ On the contrary, Whatever the power of the machine, we should also never undermine the role of human judgment or his impact on the ultimate decision its explanation.

Activity 6.4

Do you think that integrating ethical rules with emerging technologies is important? If your answer is yes, why?

What are the challenges of integrating ethical rules with the new technologies?



- ❖ What makes ethics of Technology very difficult is early integration of ethics into the technology life cycle.
 - ✓ the future is un certain.
- ❖ Ethics has traditionally not been well equipped to deal with issues of uncertainty and, in particular, future uncertainty.



6.2.1. General ethical principles

Activity 6.5

- *List down common ethical rules that must be applied in all technologies?*



6.2.1. General ethical principles

- 1) Contribute to society and to human well-being, acknowledging that all people are stakeholders in computing.
- 2) Avoid harm.
- 3) Be honest and trustworthy.
- 4) Be fair and take action not to discriminate.
- 5) Respect the work required to produce new ideas, inventions, creative works, and computing artifacts.
- 6) Respect privacy.
- 7) Honor confidentiality



6.2.2. Professional responsibilities

Activity 6.6

➤ List down ethical principles related to professional responsibility?



6.2.2. Professional responsibilities.

- 1) Strive to achieve high quality in both the processes and products of professional work.
- 2) Maintain high standards of professional competence, conduct, and ethical practice.
- 3) Know and respect existing rules pertaining to professional work.
- 4) Accept and provide appropriate professional review.
- 5) Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks.
- 6) Perform work only in areas of competence.
- 7) Foster public awareness and understanding of computing, related technologies, and their consequences.
- 8) Access computing and communication resources only when authorized or when compelled by the public good.
- 9) Design and implement systems that are robustly and usably secure.



6.2.3. Professional leadership principles

Activity 6.7

➤ *List down ethical principles related to professional leadership?*



6.2.3. principles

Professional

leadership

1. Ensure that the public good is the central concern during all professional computing work.
2. Articulate, encourage acceptance of and evaluate fulfillment of social responsibilities by members of the organization or group.
3. Manage personnel and resources to enhance the quality of working life.
4. Articulate, apply, and support policies and processes that reflect the principles of the Code.



6.2.3. Professional leadership principles

5. Create opportunities for members of the organization or group to grow as professionals.
6. Use care when modifying or retiring systems. Interface changes, the removal of features, and even software updates have an impact on the productivity of users and the quality of their work.
7. Recognize and take special care of systems that become integrated into the infrastructure of society.



6.2.3. Professional leadership principles

Activity 6.7

➤ *List down ethical principles related to professional leadership?*



6.3. Digital privacy

- What do you think about privacy in general? Is it really important?
- In this digital world, what do you mean by digital privacy? Give concrete examples?



6.3. Digital privacy

- ❖ **Digital Privacy** is the protection of personally identifiable or business identifiable information that is collected from respondents through information collection activities or from other sources.
- ❖ includes three sub-related categories;
 1. information privacy,
 2. communication privacy, and
 3. individual privacy



6.3.1. Information Privacy

Activity 6.10

➤ What do you think that private information like passwords, PIN numbers, will be guarded or shared with the public? Why?



CHAPTER 7: OTHER EMERGING TECHNOLOGIES

Content Outlines:

- ☐ Nanotechnology
- ☐ Biotechnology
- ☐ Block-chain technology
- ☐ Cloud and quantum computing
- ☐ Autonomic computing
- ☐ Computer vision
- ☐ Embedded systems,
- ☐ Cybersecurity and 3D printing.



7.1 Nanotechnology.

Definitions :

Nano-science

- As the study of phenomena and manipulation of materials at atomic, molecular and macromolecular scales, where properties differ significantly from those at a larger scale.

Nanotechnologies

- As the design, characterization, production, and application of structures, devices, and systems by controlling shape and size at the nanometer scale.



- **Nanotechnology** is science, engineering, and technology conducted at the Nano-scale, which is about 1 to 100 nanometers.
- Nano science and nanotechnology are the study and application of extremely small things and can be used across all the other science fields, such as chemistry, biology, physics, materials science, and engineering.



The ideas and concepts behind nanoscience and nanotechnology started with a talk entitled “There’s plenty of room at the bottom” by physicist

➤ **Richard Feynman** at an American Physical Society meeting at the California Institute of Technology (Caltech) on December 29, 1959, long before the term nanotechnology was used.



- He described a process in which scientists would be able to manipulate and control individual atoms and molecules.
- Over a decade later, in his explorations of ultraprecision machining,
- **Professor Norio Taniguchi** coined the term nanotechnology.

It wasn't until 1981, with the development of the scanning tunneling microscope that could "see" individual atoms, that modern nanotechnology began.



Fundamental concepts

- It's hard to imagine just how small nanotechnology is.
- One nanometer is a billionth of a meter or 10^{-9} of meters.

For example:

- ❑ There are 25,400,000 nanometers in an inch.
- ❑ A sheet of newspaper is about 100,000 nanometers thick.



On a comparative scale, if a marble were a nanometer, then one meter would be the size of the Earth.

- **Nano science and nanotechnology** involve the ability to see and to control individual atoms and molecules.
- Everything on Earth is made up of atoms—the food we eat, the clothes we wear, the buildings and houses we live in, and our own bodies.



- The microscopes needed to see things at the nanoscale were invented relatively, recently about 30 years ago.
- An atom has a diameter of about 0.1 nm. An atom's nucleus is much smaller about 0.00001 nm. Atoms are the building blocks for all matter in our universe.
- The properties of materials can be different at the nanoscale for two main reasons:



First,

- ❑ Nano-materials have a relatively larger surface area when compared to the same mass of material produced in a larger form.
- ❑ This can make materials more chemically reactive (in some cases materials that are inert in their larger form are reactive when produced in their Nano-scale form), and affect their strength or electrical properties.



Second:

- ❑ Quantum effects can begin to dominate the behavior of matter at the nanoscale particularly at the lower end – affecting the optical, electrical and magnetic behavior of materials.
- ❑ Materials can be produced that are nanoscale in one dimension (for example, nanowires, Nano rods, and nanotubes), in two dimensions (plate-like shapes like Nano coating, Nano layers, and graphene) or in all three dimensions (for example, nanoparticles).



- ❑ Today's scientists and engineers are finding a wide variety of ways to deliberately make materials at the Nano scale to take advantage of their enhanced properties such as higher strength, lighter weight, increased control of light spectrum and greater chemical reactivity than their larger-scale counterparts



7.1.3 Application of Nanotechnology

- **Medicine:** customized nanoparticles the size of molecules that can deliver drugs directly to diseased cells in your body.
- **Electronics:** it has some answers for how we might increase the capabilities of electronics devices while we reduce their weight and power consumption.
- **Food:** it has an impact on several aspects of food science, from how food is grown to how it is packaged.



- Agriculture: nanotechnology can possibly change the whole agriculture part and nourishment industry anchor from generation to preservation, handling, bundling, transportation, and even waste treatment.
- Vehicle manufacturers: Much like aviation, lighter and stronger materials will be valuable for making vehicles that are both quicker and more secure.



7.2 Biotechnology

Definition:

- **Biotechnology** is a technology based on biology.
- It harnesses cellular and biomolecular processes to develop technologies and products that help improve our lives and the health of our planet.
- We have used the biological processes of microorganisms for long time to make useful food products, such as bread and cheese, and to preserve dairy products .



- **Genetic engineering** is the process of transferring individual genes between organisms or modifying the genes in an organism to remove or add a desired trait or characteristic.
- In modern biotechnology, researchers modify DNA and proteins to shape the capabilities of living cells, plants, and animals into something useful for humans.



Application of biotechnology

- **Agriculture (Green Biotechnology):** Biotechnology had contributed a lot to modify the genes of the organism known as Genetically Modified Organisms such as Crops, Animals, Plants, Fungi, Bacteria, etc. Genetically modified crops are formed by the manipulation of DNA to introduce a new trait into the crops.



- **Medicine (Medicinal Biotechnology):** This helps in the formation of genetically modified insulin known as humulin. This helps in the treatment of a large number of diabetes patients. It has also given rise to a technique known as gene therapy. Gene therapy is a technique to remove the genetic defect in an embryo or child.
- **Aquaculture Fisheries:** It helps in improving the quality and quantity of fishes. Through biotechnology, fishes are induced to breed via gonadotropin-releasing hormone.



➤ **Environment (Environmental biotechnology):** is used in waste treatment and pollution prevention. Environmental biotechnology can more efficiently clean up many wastes than conventional methods and greatly reduce our dependence on methods for land-based disposal.



7.3. Blockchain technology

- A Blockchain is a time-stamped series of immutable records of data that is managed by a cluster of computers not owned by any single entity.
- Each of these Blocks of data (i.e. block) is secured and bound to each other using cryptographic principles (i.e. chain).
- “Blocks” on the Blockchain are made up of digital pieces of information.



Specifically, they have three parts:

1. Blocks store information about transactions like the date, time, and dollar amount of your most recent purchase from Amazon.
2. Blocks store information about who is participating in transactions. A block for your splurge purchase from Amazon would record your name along with Amazon. Instead of using your actual name, your purchase is recorded without any identifying information using a unique “digital signature,” sort of like a username.



3. Blocks store information that distinguishes them from other blocks. Much like you and I have names to distinguish us from one another, each block stores a unique code called a “hash” that allows us to tell it apart from every other block.



In order for a block to be added to the Blockchain, however, four things must happen:

1. A transaction must occur.
2. That transaction must be verified.
3. That transaction must be stored in a block.
4. That block must be given a hash.



- By design, a blockchain is resistant to modification of the data. It is "an open, distributed ledger that can record transactions between two parties efficiently and in a verifiable and permanent way".
- The blockchain network has no central authority; it is the very definition of a democratized system. Since it is a shared and immutable ledger, the information in it is open for anyone and everyone to see



The Three Pillars of Blockchain Technology

1. **Decentralization:**

- In a decentralized system the information is not stored by one single entity. In fact, everyone in the network owns the information.
- In a decentralized network ,if you wanted to interact with your friend then you can do so directly without going through a third party. That was the main ideology behind Bitcoins.



2. Transparency:

- One of the most interesting and misunderstood concepts in blockchain technology is “transparency.” Some people say that blockchain gives you privacy while some say that it is transparent.
- A person’s identity is hidden via complex cryptography and represented only by their public address. So, if you were to look up a person’s transaction history, you will not see.



3. Immutability

- Immutability, in the context of the blockchain, means that once something has been entered into the blockchain, it cannot be tampered with.
- The reason why the blockchain gets this property is that of the cryptographic hash function.



- In simple terms, hashing means taking an input string of any length and giving out an output of a fixed length. In the context of crypto currencies like bitcoin, the transactions are taken as input and run through a hashing algorithm (Bitcoin uses SHA-256) which gives an output of a fixed length.



7.3.6. Application of blockchain

A. The sharing economy

- With companies like Uber and Airbnb flourishing, the sharing economy is already a proven success .

B. Crowd funding

- Crowd funding initiatives like Kickstarter and GoFundMe are doing the advance work for the emerging peer-to-peer economy.



C. Governance

- By making the results fully transparent and publicly accessible, distributed database technology could bring full transparency to elections or any other kind of poll taking.

D. Supply chain auditing

- Consumers increasingly want to know that the ethical claims companies make about their products are real. Distributed ledgers provide an easy way to certify that the backstories of the things we buy are genuine.



E. **File storage**

- Decentralizing file storage on the internet brings clear benefits. Distributing data throughout the network protects files from getting hacked or lost.
- Interplanetary File System (IPFS) makes it easy to conceptualize how a distributed web might operate.



7.4.Cloud and quantum computing

- **Cloud computing** is a means of networking remote servers that are hosted on the Internet. Rather than storing and processing data on a local server, or a PC's hard drive, one of the following three types of cloud infrastructure is used .
- The first type is a public cloud. Here a third-party provider manages the servers, applications, and storage much like a public utility .



- A business or organization would typically use a private cloud. This might be hosted on their on-site data center, although some companies host through a third-party provider instead.
- The third option is a hybrid cloud. Here private clouds are connected to public clouds, allowing data and applications to be shared between them.



Advantages of cloud computing

- Cheap computing power.
- No need for the initial costly capital investment.
- Energy costs are shared.
- Cloud providers have vast resources of computing power at their fingertips.
- Allows you and multiple users to access your data from any location.
- Smartphone, laptop, desktop, wherever you are, you can access the data you need at any time.



7.4.3. Quantum computing

- Quantum computers truly do represent the next generation of computing. Unlike classic computers, they derive their computing power by harnessing the power of quantum physics.
- Currently, the only organization which provides a quantum computer in the cloud is IBM. They allow free access to anyone who wishes to use their 5-qubit machine. Earlier this year they installed a 17-qubit machine.
- So far over 40,000 users have taken advantage of their online service to run experiments .



Advantages of quantum computing:

- Getting a quantum computer to function usefully is an exciting prospect for scientists.
- Their gargantuan computing power would allow them to crunch very long numbers.
- They would be able to make complex calculations that would only overwhelm classic computers.



7.5. Autonomic computing (AC)

- Autonomic computing (AC) is an approach to address the complexity and evolution problems in software systems.
- It is a self-managing computing model named after, and patterned on, the human body's autonomic nervous system.
- The goal of autonomic computing is to create systems that run themselves, capable of high-level functioning while keeping the system's complexity invisible to the user.



Characteristics of Autonomic Systems

- **An autonomic system** can self-configure at runtime to meet changing operating environments, self-tune to optimize its performance, self-heal when it encounters unexpected obstacles during its operation, and of particular current interest.
- Autonomic systems/applications exhibit eight defining characteristics:



- **Self-Awareness:** An autonomic application/system “knows itself” and is aware of its state and its behaviors.
- **Self-Configuring:** An autonomic application/system should be able to configure and reconfigure itself under varying and unpredictable conditions.
- **Self-Optimizing:** An autonomic application/system should be able to detect suboptimal behaviors and optimize itself to improve its execution.



- **Self-Healing:** An autonomic application/system should be able to detect and recover from potential problems and continue to function smoothly.
- **Self-Protecting:** An autonomic application/system should be capable of detecting and protecting its resources from both internal and external attacks and maintaining overall system security and integrity.



- **Context-Aware:** An autonomic application/system should be aware of its execution environment and be able to react to changes in the environment.
- **Open:** An autonomic application/system must function in a heterogeneous world and should be portable across multiple hardware and software architectures. Consequently, it must be built on standard and open protocols and interfaces.
- **Anticipatory:** An autonomic application/system should be able to anticipate to the extent possible, its needs and behaviors and those of its context, and be able to manage itself proactively.



7.6.Computer vision

Definition :

- It is an interdisciplinary scientific field that deals with how computers can be made to gain a high-level understanding of digital images or videos.
- Computer vision tasks include methods for acquiring, processing, analyzing and understanding digital images, and extraction of high-dimensional data from the real world in order to produce numerical or symbolic information, e.g. in the forms of decisions



- 1. Acquiring an image:** Images, even large sets, can be acquired in real-time through video, photos or 3D technology for analysis.
- 2. Processing the image:** Deep learning models automate much of this process, but the models are often trained by first being fed thousands of labeled or pre-identified images.
- 3. Understanding the image:** The final step is the interpretative step, where an object is identified or classified.



There are many types of computer vision that are used in different ways:

- Image segmentation partitions an image into multiple regions or pieces to be examined separately.
- Object detection identifies a specific object in an image. Advanced object detection recognizes many objects in a single image: a football field, an offensive player, a defensive player, a ball and so on. These models use an X, Y coordinate to create a bounding box and identify everything inside the box.



- Facial recognition is an advanced type of object detection that not only recognizes a human face in an image but identifies a specific individual.
- Edge detection is a technique used to identify the outside edge of an object or landscape to better identify what is in the image.
- Pattern detection is a process of recognizing repeated shapes, colors and other visual indicators in images.



- Image classification groups images into different categories.
- Feature matching is a type of pattern detection that matches similarities in images to help classify them.



Applications of computer vision

- Computer vision is being used today in a wide variety of real-world applications, which include:
 - ❑ **Optical character recognition** (OCR): reading handwritten postal codes on letters and automatic number plate recognition (ANPR);
 - ❑ **Machine inspection**: rapid parts inspection for quality assurance using stereo vision with specialized illumination to measure tolerances on aircraft wings or auto body parts or looking for defects in steel castings using X-ray vision;



- ❑ **Retail:** object recognition for automated checkout lanes
- ❑ **Medical** imaging: registering pre-operative and intra-operative imagery or performing long-term studies of people's brain morphology as they age;
- ❑ **Automotive safety:** detecting unexpected obstacles such as pedestrians on the street, under conditions where active vision techniques such as radar or lidar do not work well.
- ❑ **Surveillance:** monitoring for intruders, analyzing highway traffic and monitoring pools for drowning victims;
- ❑ **Fingerprint recognition and biometrics:** for automatic access authentication as well as forensic applications.



7.7.Embedded systems

- It is a controller with a dedicated function within a larger mechanical or electrical system, often with real-time computing constraints. It is *embedded* as part of a complete device often including hardware and mechanical parts.
- Modern embedded systems are often based on microcontrollers (i.e. microprocessors with integrated memory and peripheral interfaces), but ordinary microprocessors (using external chips for memory and peripheral interface circuits) are also common, especially in more complex systems.



Advantages and disadvantages of embedded system

Advantages of Embedded:

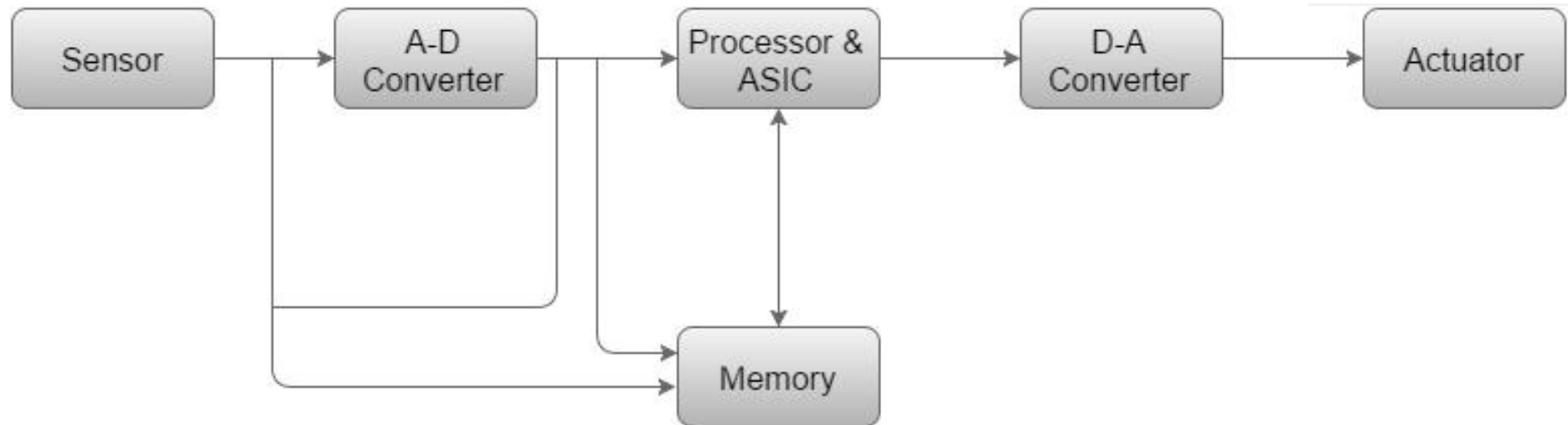
- ☐ Easily Customizable.
- ☐ Low power consumption.
- ☐ Low cost.
- ☐ Enhanced performance

Disadvantages of Embedded systems:

- ☐ High development effort.
- ☐ Larger time to market

Basic Structure of an Embedded System:

- **Sensor** – It measures the physical quantity and converts it to an electrical signal which can be read by an observer or by any electronic instrument like an A2D converter. A sensor stores the measured quantity to the memory.
- **A-D Converter** – An analog-to-digital converter converts the analog signal sent by the sensor into a digital signal.





- **Processor & ASICs** – Processors process the data to measure the output and store it to the memory.
- **D-A Converter** – A digital-to-analog converter converts the digital data fed by the processor to analog data.
- **Actuator** – An actuator compares the output given by the D-A Converter to the actual (expected) output stored in it and stores the approved output.



7.8.Cybersecurity

Definition:

- It is the protection of computer systems from the theft of or damage to their hardware, software, or electronic data, as well as from the disruption or misdirection of the services they provide.
- Cyber security is becoming more important due to
 - ❑ Increased reliance on computer systems.
 - ❑ The Internet and wireless network standards such as Bluetooth and Wi-Fi, and due to the growth of smart devices, including smartphones, televisions, and the various devices that constitute the Internet of Things.



- **Cybercrime** is any criminal activity that involves a computer, network devices or a network.
- It is often committed by the people of destructive and criminal mindset either for revenge, greed or adventure.



Cyber security measures

- Some of cyber security measures to be taken to prevent cybercrime .
 - ❑ **Staff awareness training:** - Human error is the leading cause of data breaches, so you need to equip staff with the knowledge to deal with the threats they face.
 - ❑ **Application security:** - Web application vulnerabilities are a common point of intrusion for cybercriminals. As applications play an increasingly critical role in business, it is vital to focus on web application security.



- **Network security:** - is the process of protecting the usability and integrity of network and data.

This is achieved by conducting a network penetration test, which scans your network for vulnerabilities and security issues .

- **Leadership commitment:** - is the key to cyber resilience. Without it, it is very difficult to establish or enforce effective processes .Top management must be prepared to invest in appropriate cybersecurity resources, such as awareness training.

- **Password management :-**provides guidance to ensure that individuals or organizations create strong passwords and keep them secure .



Types of cyber security threats

- **Ransomware:** - It is a type of malicious software. It is designed to extort money by blocking access to files or the computer system until the ransom is paid. Paying the ransom does not guarantee that the files will be recovered or the system restored.
- **Malware:-** it is a type of software designed to gain unauthorized access or to cause damage to a computer.



- **Social engineering:** - it is a tactic that adversaries use to trick you into revealing sensitive information. They can solicit a monetary payment or gain access to your confidential data.
- **Phishing:** - it is the practice of sending fraudulent emails that resemble emails from reputable sources. The aim is to steal sensitive data like credit card numbers and login information. It's the most common type of cyber-attack.



Benefits of utilizing cybersecurity include:

- Business protection against malware, ransomware, phishing, and social engineering.
- Protection for data and networks.
- Prevention of unauthorized users.
- Improves recovery time after a breach.
- Protection for end-users.
- Improved confidence in the product for both developers and customers.



Cyber security vendors

- Vendors in cybersecurity fields will typically use endpoint, network and advanced threat protection security as well as data loss prevention.
- Three commonly known cybersecurity vendors include Cisco, McAfee, and Trend Micro.



7.9. Additive manufacturing (3D Printing)

Let's start by considering what the experts have to say about AM (3D printing)

- The American Society for Testing and Materials (ASTM) equates the two terms in their definition: “Additive manufacturing, also known as 3D printing, uses computer-aided design to build objects layer by layer.”
- Wikipedia says, “Today, the precision, repeatability, and material range has increased to the point that 3D printing is considered as industrial production technology, with the name of additive manufacturing.”
- Author Deve Turbide puts it simply, suggesting that additive manufacturing is “the industrial version of 3D printing”.



3D Printing: It's All About the Printer

- When MIT invented binder jet printing in the 1980s, they wrote: “three-dimensional printing is a process under development at MIT for the rapid and flexible production of prototype parts, end-use parts, and tools directly from a CAD model.”
- Today our concept of “3D printing” is much broader, but the term is often associated with filament-based plastic printers .But there are also binder jet printers, laser metal 3D printers, as well as glass and clay 3D printers.



Additive Manufacturing:

- “Additive manufacturing” (AM) is a big-picture term more at home in the boardroom than the factory floor or garage. Naturally, AM separates itself from older, subtractive technologies like milling.
- Additive manufacturing (AM) describes types of advanced manufacturing that are used to create three-dimensional structures out of plastics, metals, polymers and other materials that can be sprayed through a nozzle or aggregated in a vat .



Thank you